Path Integrals in Quantum Mechanics

Lecture notes, Parts I and II

Typeset from handwritten notes (pp. 1–42), corrected, reorganized and expanded

October 2026

Preface

These notes are a typeset version of handwritten lecture notes on path integrals in non-relativistic quantum mechanics (handwritten pp. 1–42). Three kinds of changes were made.

1. **Corrections.** Mathematical slips, sign errors, missing terms and imprecise statements have been fixed in place. A complete log is in Appendix B.
2. **Reordering.** The handwritten notes introduced two tools *after* they were first needed: canonical transformations (Additional Note 1, hw. pp. 7–11), which motivate Dirac's proposal, and distributions and functional derivatives (Additional Note 2, hw. pp. 27–31), which are used on hw. pp. 19 and 24–25. Both now come first (Sections 2–3). The thermal partition function has been moved into the chapter on complex time, where it belongs conceptually. The overall arc of the original is kept.
3. **Additions**, marked ★ in headings: a formal treatment of the stationary-phase method; a check that the path integral reproduces the Schrödinger equation; Green's functions in energy space; the general Schwinger–Dyson equation and contact terms; a chapter of intuition for complex time (Section 8); the Maslov phase and a spectral check for the harmonic oscillator.

Tags such as [hw. p. 12] point back to the handwritten page. Appendix B maps every handwritten page to its section here.

Conventions used throughout

- $\hbar$ is kept explicit. $T \equiv t_f - t_i > 0$ is the elapsed time; the time-ordering operator is written $\mathcal{T}$ to avoid a clash with T.
- Heisenberg position eigenstates: $|x,t\rangle \equiv e^{i\hat{H}t/\hbar}|x\rangle$, so that $\hat{x}(t)|x,t\rangle = x|x,t\rangle$. The propagator is $K(f;i) \equiv \langle x_f,t_f|x_i,t_i\rangle = \langle x_f|e^{-i\hat{H}T/\hbar}|x_i\rangle$.
- Time slicing: $\varepsilon = T/N$, $t_n = t_i + n\varepsilon$, $x_n \equiv x(t_n)$, with $x_0 = x_i$ and $x_N = x_f$ fixed.
- The "$i\varepsilon$" regulator is written $\eta \to 0^+$, so that it does not clash with the time step ε.
- Euclidean (imaginary) time u is defined by $t = -iu$.
- $\langle x|p\rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$, $\int \mathrm{d}x\,|x\rangle\langle x| = \int \mathrm{d}p\,|p\rangle\langle p| = \mathbb{1}$.
- Diagrams of paths in spacetime have time running *upward*. Plots of functions (mode shapes, the polymer picture) have time running to the right.

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1 Motivation and overview

1.1 Two ways to formulate mechanics [hw. p. 1]

In Schrödinger's formulation the state obeys a first-order equation in time,

$$i\hbar\,\partial_t|\psi(t)\rangle = \hat{H}|\psi(t)\rangle, \tag{1.1}$$

and the basic problem is an initial-value problem: given $|\psi(t_i)\rangle$, find $|\psi(t_f)\rangle$. Time plays a privileged role. It is the evolution parameter, while position is an operator. This is the quantum analogue of the *Hamiltonian* formulation of classical mechanics.

Classical mechanics has a second formulation, global in time rather than step-by-step. One assigns to every path $x(t)$ with fixed endpoints a number, the *action*

$$S[x] = \int_{t_i}^{t_f} \mathrm{d}t\, L(x, \dot{x}, t), \qquad x(t_i) = x_i, \quad x(t_f) = x_f, \tag{1.2}$$

and the physical trajectory is the one that makes S stationary. In relativistic field theory the action is $S = \int \mathrm{d}^4x\, \mathcal{L}$ with $\mathcal{L}$ a Lorentz scalar, so this formulation is the natural home of relativistic symmetry. Dirac (1933) asked whether quantum mechanics can be built on the action instead of the Hamiltonian. Feynman (1948) completed the answer:

$$\boxed{\langle x_f, t_f | x_i, t_i\rangle = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x\; e^{iS[x]/\hbar}} \tag{1.3}$$

The transition amplitude is a sum over *all* paths, each weighted by a pure phase built from its classical action.

Remark 1.1 (An honest caveat about "equal footing"). For a single non-relativistic particle the path integral does not yet treat space and time alike. Time remains a parameter and only $x(t)$ is integrated over. The payoff of Dirac's programme appears in two places. In field theory, $\int \mathcal{D}\phi\, e^{iS[\phi]/\hbar}$ with a Lorentz-invariant S makes the symmetry manifest. In the worldline formalism of a relativistic particle, time is promoted to a dynamical variable $x^0(\tau)$. Both are treated in the later parts of the course.

1.2 What the path integral buys us

- **A manifest classical limit.** As $\hbar \to 0$ the integral is dominated by the stationary point $\delta S = 0$, which is classical mechanics (Section 4.6).
- **Visible symmetries.** Symmetries of S become symmetries of the quantum theory with almost no operator algebra, which is why the path integral is the standard language of gauge theories.

- **A direct route to field theory**, by replacing $x(t)$ with $\phi(\mathbf{x}, t)$.
- **A bridge to statistical mechanics.** In imaginary time the weight $e^{iS/\hbar}$ becomes a Boltzmann weight $e^{-S_{\mathrm{E}}/\hbar}$ (Section 8).
- **Non-perturbative tools**, such as instantons and tunnelling, which are invisible in operator perturbation theory.

1.3 Plan of the notes

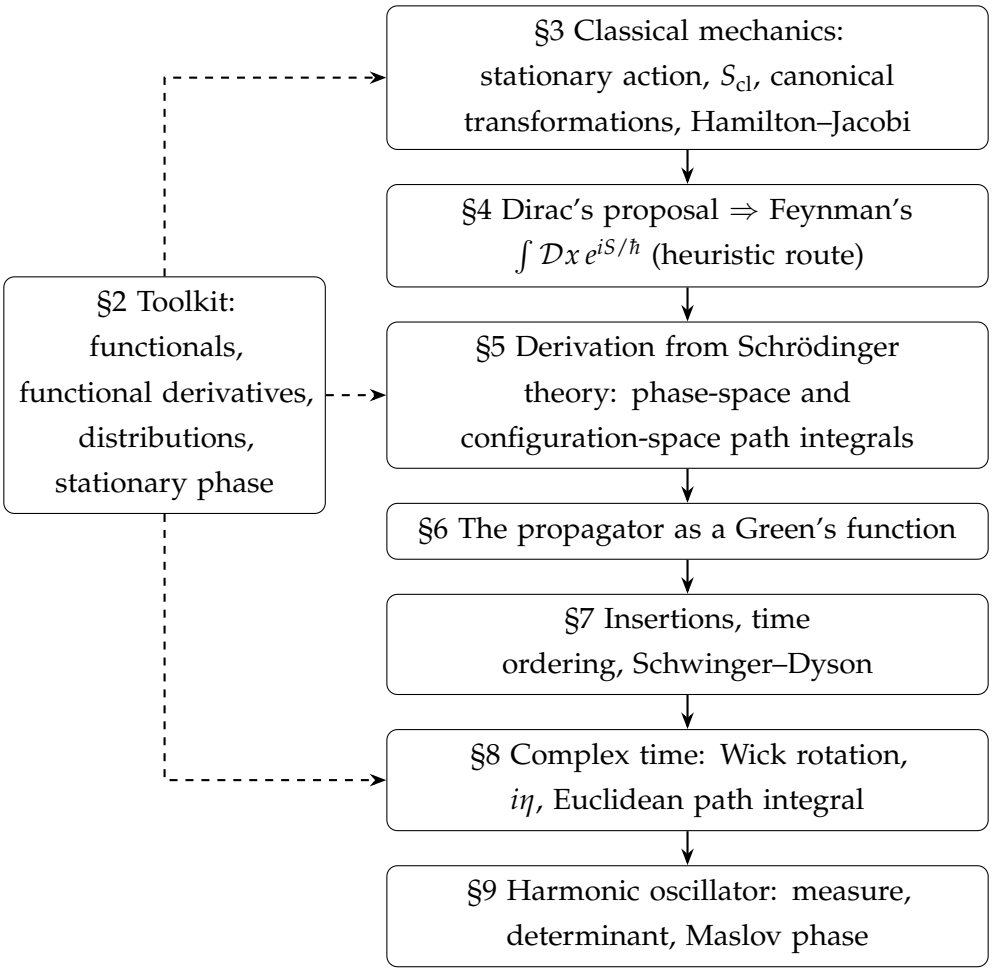


Figure 1: Logical structure. The toolkit of Section 2 feeds every later chapter. Sections 4 and 5 reach the same formula by two routes: Dirac's heuristic and an honest derivation.

2 Mathematical toolkit

This section collects the tools that the rest of the notes use repeatedly. In the hand-written notes they were developed in Additional Note 2 [hw. pp. 27–31], after their first use.

2.1 Functions versus functionals [hw. p. 2]

A function $f : \mathbb{R} \to \mathbb{R}$ assigns a number to a *point* x. A functional assigns a number to an entire *function*.

Definition 2.1. Let $\mathcal{F}_{if}$ be the set of continuous paths $x : [t_i, t_f] \to \mathbb{R}$ with $x(t_i) = x_i$ and $x(t_f) = x_f$. A *functional* is a map $F : \mathcal{F}_{if} \to \mathbb{C}$, written $F[x]$. The square bracket signals that the argument is the whole path, not its value at one time.

The action $S[x] = \int L \, \mathrm{d}t$ is the central example. Two structural facts are worth stating precisely.

- $\mathcal{F}_{if}$ is *not* a vector space, because the sum of two paths has the wrong endpoints. It is an *affine* space: any path can be written $x = \bar{x} + \eta$, where $\bar{x}$ is a fixed reference path and η lies in the vector space $\mathcal{V} = \{\eta : \eta(t_i) = \eta(t_f) = 0\}$ of *variations* (or *fluctuations*).
- $\mathcal{V}$ is infinite-dimensional. The value $\eta(t)$ at each instant is an independent direction.

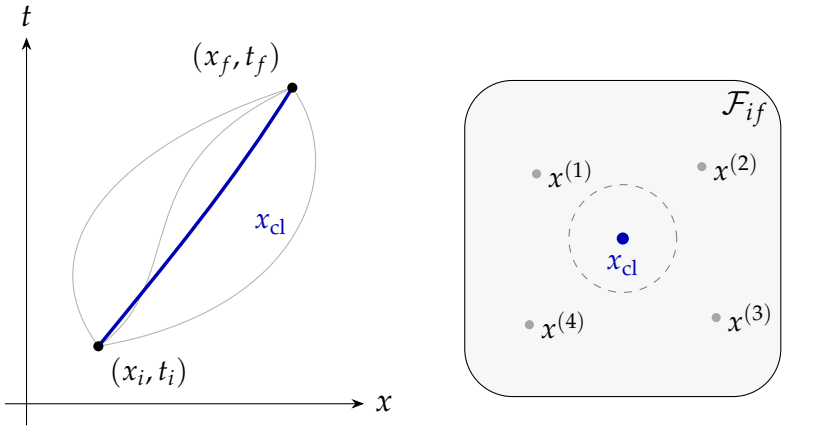


Figure 2: Left: several paths with the same endpoints; the classical path is the thick (blue) curve. Right: the same paths as points of the infinite-dimensional space $\mathcal{F}_{if}$. The dashed circle is a neighbourhood of x_{cl}, the paths that interfere constructively in the classical limit (Section 4.6).

2.2 Functionals as limits of ordinary functions [hw. p. 29]

Slice the time interval into N steps of size $\varepsilon = T/N$ and record the path only at $t_n = t_i + n\varepsilon$. A path becomes the finite list $(x_1, \ldots, x_{N-1})$, since $x_0 = x_i$ and $x_N = x_f$ are fixed. A functional becomes an ordinary function of $N-1$ variables, $F[x] \to F_N(x_1, \ldots, x_{N-1})$. Integration over paths is *defined* as the limit of ordinary multiple integrals:

$$\int \mathcal{D}x\, F[x] \equiv \lim_{N\to\infty} \mathcal{N}_N \int \prod_{n=1}^{N-1} \mathrm{d}x_n\, F_N(x_1, \ldots, x_{N-1}). \tag{2.1}$$

The normalization $\mathcal{N}_N$ is not yet determined. It is fixed in Section 5.3, where we find $\mathcal{N}_N = A^N$ with $A = \sqrt{m/2\pi i\hbar\varepsilon}$. Multivariable calculus becomes functional calculus in the continuum limit:

multivariable calculus	$\xrightarrow{\varepsilon\to 0}$	functional calculus
$\int \prod_n \mathrm{d}x_n$		$\int \mathcal{D}x$
$\frac{1}{\varepsilon}\frac{\partial}{\partial x_n}$		$\frac{\delta}{\delta x(t_n)}$
$\frac{\delta_{nn'}}{\varepsilon}$		$\delta(t_n - t_{n'})$

The factors of ε in this dictionary are essential and are explained next.

2.3 The functional derivative [hw. pp. 29–31]

Definition 2.2. The functional derivative $\delta F/\delta x(t)$ is defined by the first-order variation

$$\delta F \equiv F[x+\delta x] - F[x] = \int_{t_i}^{t_f} \mathrm{d}t\, \frac{\delta F}{\delta x(t)}\, \delta x(t) + O(\delta x^2), \tag{2.2}$$

required for every variation $\delta x \in \mathcal{V}$.

Proposition 2.3 (Discrete form)**.** *For $t = t_n$ an interior slice time,*

$$\frac{\delta F}{\delta x(t_n)} = \lim_{\varepsilon\to 0} \frac{1}{\varepsilon} \frac{\partial F_N}{\partial x_n}. \tag{2.3}$$

Proof. For the discretized functional, $\delta F_N = \sum_n \frac{\partial F_N}{\partial x_n}\delta x_n = \sum_n \varepsilon \left[\frac{1}{\varepsilon}\frac{\partial F_N}{\partial x_n}\right]\delta x_n$. The right-hand side is a Riemann sum, which tends to $\int \mathrm{d}t\, \frac{\delta F}{\delta x(t)}\delta x(t)$ precisely when the bracket is identified with $\delta F/\delta x(t_n)$. □

The $1/\varepsilon$ is what turns the Kronecker delta into a Dirac delta. Applying (2.3) to $F = x(t_{n'}) = x_{n'}$ gives $\delta_{nn'}/\varepsilon$, which tends to $\delta(t_n - t_{n'})$:

$$\boxed{\frac{\delta x(t')}{\delta x(t)} = \delta(t - t')} \tag{2.4}$$

Without the $1/\varepsilon$ the continuum limit of $\delta_{nn'}$ would be meaningless. The functional derivative obeys the usual rules (linearity, Leibniz, chain rule), so for example

$$\frac{\delta}{\delta x(t)} \int \mathrm{d}t'\, f\big(x(t')\big) = f'\big(x(t)\big). \tag{2.5}$$

Example 2.4 (Harmonic-oscillator action)**.** Let $S[x] = \int_{t_i}^{t_f} \mathrm{d}t'\, \frac{1}{2} m\big(\dot{x}(t')^2 - \omega^2 x(t')^2\big)$. Using $\delta\dot{x} = \frac{\mathrm{d}}{\mathrm{d}t'}\delta x$,

$$\begin{aligned} \delta S &= \int \mathrm{d}t'\, m\big[\dot{x}(t')\,\delta\dot{x}(t') - \omega^2 x(t')\,\delta x(t')\big] \\ &= \Big[m\dot{x}\,\delta x\Big]_{t_i}^{t_f} - \int \mathrm{d}t'\, m\big[\ddot{x}(t') + \omega^2 x(t')\big]\delta x(t'). \end{aligned} \tag{2.6}$$

The boundary term vanishes because $\delta x(t_i) = \delta x(t_f) = 0$. Comparing with (2.2),

$$\frac{\delta S}{\delta x(t)} = -m\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)x(t), \qquad t_i < t < t_f. \tag{2.7}$$

Remark 2.5. The functional derivative is defined only at interior times. The endpoint values are fixed, so there is nothing to vary there.

2.4 Distributions [hw. pp. 19, 27–29]

Definition 2.6. The space of *test functions* is $\mathcal{D} = C_c^\infty(\mathbb{R})$: smooth functions that vanish outside a bounded interval. A *distribution* is a continuous linear map $\mathsf{T} : \mathcal{D} \to \mathbb{C}$, written $\varphi \mapsto \langle \mathsf{T}, \varphi \rangle$.

Compact support matters. It is what makes every boundary term in an integration by parts vanish. Merely smooth test functions would not do, and with them one would have to carry and cancel terms like $f(\infty)$, as on [hw. p. 19].

Ordinary functions are distributions. A locally integrable f defines $\langle \mathsf{T}_f, \varphi \rangle = \int \mathrm{d}x\, f(x)\varphi(x)$. By the fundamental lemma of the calculus of variations, $\mathsf{T}_f = \mathsf{T}_g$ implies $f = g$ almost everywhere. So describing an ordinary function by all its integrals against test functions *loses nothing*.[1] The point of distributions is that this description makes sense for *more* objects. Some distributions are not of the form T_f for any function f. The prime example is the Dirac delta, $\langle \delta, \varphi \rangle = \varphi(0)$. The handwritten remark that "we evaluate f at the point φ of C^∞" is exactly the right picture: a distribution is a function on the space of test functions.

[1] This corrects the statement on hw. p. 27 that "the reverse is not true".

Derivatives. Integration by parts with no boundary terms motivates

$$\langle \mathsf{T}', \varphi \rangle \equiv -\langle \mathsf{T}, \varphi' \rangle. \tag{2.8}$$

For the step function $\theta(x)$ (equal to 1 for $x > 0$ and 0 for $x < 0$; its value at 0 is irrelevant), $\langle \theta', \varphi \rangle = -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0)$, so $\boxed{\theta' = \delta}$. Similarly $\langle \delta', \varphi \rangle = -\varphi'(0)$. Multiplication by a smooth function g is defined by $\langle g\mathsf{T}, \varphi \rangle = \langle \mathsf{T}, g\varphi \rangle$.

Proposition 2.7 (Product rules for δ and δ')**.** *For smooth g, as distributions in t at fixed t':*

$$g(t)\,\delta(t-t') = g(t')\,\delta(t-t'), \tag{2.9}$$

$$g(t)\,\partial_t\delta(t-t') = g(t')\,\partial_t\delta(t-t') - g'(t')\,\delta(t-t'). \tag{2.10}$$

Equivalently, in the form used in the handwritten notes,

$$\partial_t\delta(t-t')\,g(t') = \partial_t\delta(t-t')\,g(t) + \delta(t-t')\,\partial_t g(t). \tag{2.11}$$

Proof. Test against $\varphi(t)$ and integrate over t. For (2.10):

$$\text{LHS:}\quad \langle \partial_t\delta(\cdot - t'), g\varphi \rangle = -(g\varphi)'(t') = -g'(t')\varphi(t') - g(t')\varphi'(t'), \tag{2.12}$$

$$\text{RHS:}\quad g(t')\big(-\varphi'(t')\big) - g'(t')\varphi(t'). \tag{2.13}$$

These agree for every φ. Equation (2.11) follows by rearranging and using (2.9) on the g' term. □

> **Caution: test against the right variable**
>
> A distributional identity in t must be tested against an *arbitrary* $\varphi(t)$ and integrated over t. Multiplying by $\varphi(t')$ and integrating over t (the check chosen on hw. p. 29) tests only against a function that is constant in the integration variable. It verifies just that both sides have the same integral over t. That check would "prove" the false identity $\partial_t\delta(t-t') = 0$, since $\int \mathrm{d}t\,\partial_t\delta(t-t') = 0$. The proof above uses the correct test.

Remark 2.8. The proof uses only linearity and differentiability in t. The identities therefore hold verbatim when g is an operator-valued function such as $g(t) = [\hat{x}(t), \hat{x}(t')]$, inside any matrix element. This is how they are used in Section 7.4.

2.5 Oscillatory integrals and stationary phase★

Path integrals are built from integrals of pure phases. Two facts about such integrals carry most of the weight in what follows.

2.5.1 The Fresnel integral, by rotating the contour

Proposition 2.9. *For real $a \neq 0$,*

$$\int_{-\infty}^{\infty} \mathrm{d}x\, e^{iax^2} = \sqrt{\frac{\pi}{|a|}}\, e^{i\,\mathrm{sgn}(a)\,\pi/4} \equiv \sqrt{\frac{i\pi}{a}}. \tag{2.14}$$

Proof. Take $a > 0$ (the case $a < 0$ is the complex conjugate). The integrand does not decay on the real axis, but e^{iaz^2} is an entire function of z. Integrate it around the wedge of Fig. 3: out along the real axis to R, along the arc $|z| = R$ to angle $\pi/4$, and back to the origin along the ray $z = se^{i\pi/4}$. There are no singularities inside, so Cauchy's theorem gives a total of zero.

- On the arc, $z = Re^{i\phi}$ and $|e^{iaz^2}| = e^{-aR^2 \sin 2\phi}$. Since $\sin 2\phi \geq 4\phi/\pi$ on $[0, \pi/4]$, the arc contributes at most $\int_0^{\pi/4} e^{-4aR^2\phi/\pi} R\,\mathrm{d}\phi \leq \pi/(4aR) \to 0$.
- On the ray, $z^2 = is^2$, so $e^{iaz^2} = e^{-as^2}$. The integrand is a decaying Gaussian.

Hence $\int_0^{\infty} e^{iax^2}\mathrm{d}x = e^{i\pi/4}\int_0^{\infty} e^{-as^2}\mathrm{d}s = \frac{1}{2}e^{i\pi/4}\sqrt{\pi/a}$. Doubling for the full line gives (2.14). □

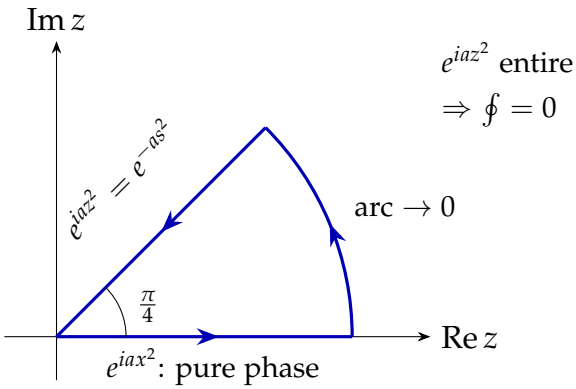


Figure 3: Rotating the contour by $\pi/4$ turns the non-decaying phase e^{iax^2} into a decaying Gaussian, and Cauchy's theorem guarantees the same answer. This is the prototype of the Wick rotation of Section 8, where time instead of x is rotated.

Two corollaries are used repeatedly. Both follow from (2.14), by completing the square and by differentiating with respect to a:

$$\int \mathrm{d}x\, e^{iax^2+ibx} = \sqrt{\frac{i\pi}{a}}\, e^{-ib^2/4a}, \qquad \int \mathrm{d}x\, x^2 e^{iax^2} = \frac{i}{2a}\sqrt{\frac{i\pi}{a}}. \tag{2.15}$$

Remark 2.10. There is a second way to make sense of (2.14): replace $a \to a + i\eta$ with $\eta \to 0^+$, so that the integrand acquires a damping factor $e^{-\eta x^2}$. Rotating the contour and adding infinitesimal damping give the same answer. These are exactly the two methods of Section 8, the Wick rotation and the $i\eta$ prescription, in miniature.

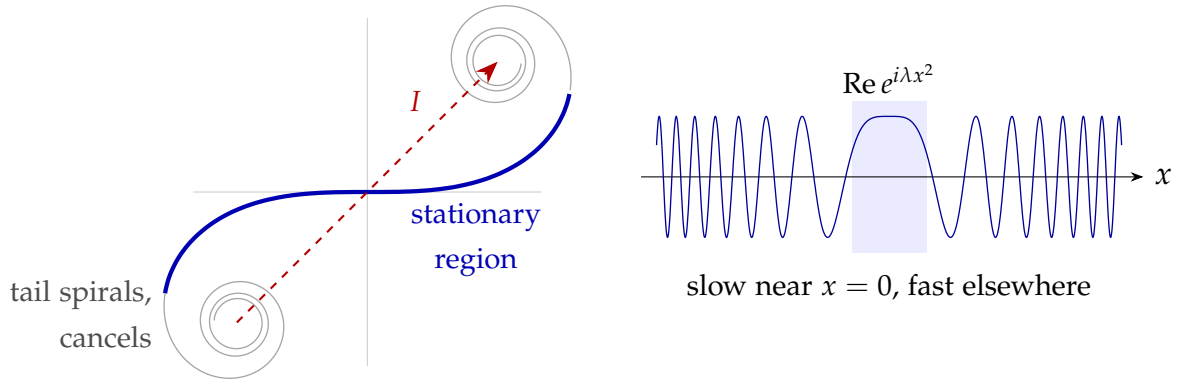


Figure 4: Stationary phase for $f(x) = x^2$. Left: the phasor chain $\int^x e^{i\pi s^2/2} \mathrm{d}s$ (Cornu spiral). The thick blue segment comes from the neighbourhood of the stationary point; the gray tails curl up and cancel. The dashed red arrow is the full integral. Right: the real part of the integrand oscillates slowly only near the stationary point (shaded).

2.5.2 The method of stationary phase

Consider $I(\lambda) = \int \mathrm{d}x\, g(x)\, e^{i\lambda f(x)}$ with f real and $\lambda \to \infty$.

- **Non-stationary regions cancel.** Where $f' \neq 0$, write $e^{i\lambda f} = \frac{1}{i\lambda f'} \frac{\mathrm{d}}{\mathrm{d}x} e^{i\lambda f}$ and integrate by parts. Each repetition gains a factor $1/\lambda$, so such regions contribute less than any power of $1/\lambda$.
- **Stationary points survive.** Near x_0 with $f'(x_0) = 0$ and $f''(x_0) \neq 0$, expand $f \approx f(x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2$ and use (2.14):

$$\boxed{I(\lambda) \simeq \sum_{\text{stat. pts}} g(x_0) \sqrt{\frac{2\pi}{\lambda |f''(x_0)|}}\, e^{i\lambda f(x_0) + i\, \operatorname{sgn} f''(x_0)\, \pi/4}} \tag{2.16}$$

In n dimensions, with Hessian $H_{jk} = \partial_j \partial_k f(\mathbf{x}_0)$ and signature $\sigma(H)$ (the number of positive minus negative eigenvalues),

$$I(\lambda) \simeq g(\mathbf{x}_0) \Big(\frac{2\pi}{\lambda}\Big)^{n/2} |\det H|^{-1/2}\, e^{i\lambda f(\mathbf{x}_0) + i\pi\sigma(H)/4}. \tag{2.17}$$

The phase $e^{\pm i\pi/4}$ per direction is the *Maslov phase*. It records whether the stationary point is a minimum or a maximum along that direction, and it reappears for the harmonic oscillator in Section 9.6.

Remark 2.11. If f is *exactly* quadratic, the expansion has no higher terms and (2.16) is exact, not asymptotic. This is why the free particle and the harmonic oscillator can be solved exactly by stationary phase (Section 9).

> **Intuition: the phasor chain**
>
> Picture the integral as a chain of tiny arrows, one per point x, each of length $\mathrm{d}x$ and pointing in the direction $\lambda f(x)$. Where the phase changes quickly, consecutive

arrows turn sharply and the chain curls up into a tight spiral with almost no net displacement. Near a stationary point the direction hardly changes, so the arrows line up and add. For $f = x^2$ the chain is the Cornu (Euler) spiral of Fig. 4. The resultant arrow joining its two spiral centres is the value of the integral, and it is built almost entirely from the nearly straight central stretch.

3 Classical preliminaries

Dirac's proposal (Section 4) rests on one classical fact: *the action generates the canonical transformation that evolves a system in time*. This section develops that fact. In the handwritten notes it appeared as Additional Note 1 [hw. pp. 7–11], after the proposal it justifies.

3.1 Stationary (not least) action [hw. p. 2]

For $S[x] = \int_{t_i}^{t_f} L(x, \dot{x}, t)\,\mathrm{d}t$ and a variation $\delta x \in \mathcal{V}$,

$$\delta S = \int_{t_i}^{t_f} \mathrm{d}t \left[\frac{\partial L}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}}\right]\delta x + \left[\frac{\partial L}{\partial \dot{x}}\delta x\right]_{t_i}^{t_f}. \tag{3.1}$$

The boundary term vanishes, so $\delta S/\delta x(t)$ is the Euler–Lagrange expression, and the classical path is characterized by

$$\frac{\delta S}{\delta x(t)}\Big|_{x_{\mathrm{cl}}} = 0 \quad \Longleftrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0. \tag{3.2}$$

> **Caution: "least" action is usually false**
>
> The classical path makes S *stationary*. It need not be a minimum. For the harmonic oscillator, expand a variation as $\eta(t) = \sum_{n\geq 1} a_n \sin\frac{n\pi(t-t_i)}{T}$. The action is quadratic and the linear term vanishes, so the change of the action is exactly the second-order term (written $\frac{1}{2}\delta^2 S$, as in Section 4.6):
>
> $$S[x_{\mathrm{cl}} + \eta] - S_{\mathrm{cl}} = \tfrac{1}{2}\,\delta^2 S[\eta] = \frac{m}{2}\int_{t_i}^{t_f} (\dot{\eta}^2 - \omega^2\eta^2)\mathrm{d}t = \frac{mT}{4}\sum_{n\geq 1} a_n^2\Big[\Big(\frac{n\pi}{T}\Big)^2 - \omega^2\Big]. \tag{3.3}$$
>
> This is positive for every η only if $\omega T < \pi$. For $\omega T > \pi$ there are $\lfloor \omega T/\pi \rfloor$ directions in which S *decreases*, and the classical path is a saddle. The threshold $\omega T = \pi$ is a *conjugate (focal) point*: every classical path leaving x_i refocuses at $-x_i$. Stationary phase only ever uses $\delta S = 0$, so "stationary" is the right word. The negative directions return as the Maslov phase in Section 9.6.

3.2 The action functional versus Hamilton's principal function

Two different objects share the letter S, and keeping them apart is essential.

$S[x]$ **versus** S_{cl}
• The **action functional** $S[x] = \int L\,\mathrm{d}t$ is defined on *every* path in $\mathcal{F}_{if}$, classical or not. This is the object inside $\int \mathcal{D}x\, e^{iS[x]/\hbar}$.

- **Hamilton's principal function** is the action evaluated on the classical path,

$$S_{\mathrm{cl}}(x_f, t_f; x_i, t_i) \equiv S[x_{\mathrm{cl}}]. \tag{3.4}$$

  It is an ordinary *function* of the endpoints, not a functional. (We assume a unique classical path. When there are several, there are several branches of S_{cl}.)

Only S_{cl} generates canonical transformations.

Proposition 3.1.

$$\frac{\partial S_{\mathrm{cl}}}{\partial x_f} = p_f, \qquad \frac{\partial S_{\mathrm{cl}}}{\partial x_i} = -p_i, \qquad \frac{\partial S_{\mathrm{cl}}}{\partial t_f} = -H(x_f, p_f), \qquad \frac{\partial S_{\mathrm{cl}}}{\partial t_i} = +H(x_i, p_i), \tag{3.5}$$

where $p = \partial L/\partial \dot{x}$ is evaluated on the classical path.

Proof. Vary the endpoints. Because x_{cl} satisfies the Euler–Lagrange equation, only the boundary term survives: $\delta S_{\mathrm{cl}} = p_f\,\delta x_f - p_i\,\delta x_i$. This gives the first two relations. For the time derivative, along the trajectory $\frac{\mathrm{d}S_{\mathrm{cl}}}{\mathrm{d}t_f} = L = \frac{\partial S_{\mathrm{cl}}}{\partial t_f} + p_f \dot{x}_f$, hence $\partial S_{\mathrm{cl}}/\partial t_f = L - p\dot{x} = -H$. The last relation follows in the same way. □

Example 3.2 (Free particle)**.** x_{cl} is uniform motion and $S_{\mathrm{cl}} = \frac{m(x_f - x_i)^2}{2T}$. Then $\partial S_{\mathrm{cl}}/\partial x_f = m(x_f - x_i)/T = p$ and $\partial S_{\mathrm{cl}}/\partial t_f = -\frac{m(x_f - x_i)^2}{2T^2} = -\frac{p^2}{2m} = -H$, as (3.5) requires.

3.3 Canonical transformations and generating functions [hw. pp. 7–9]

Definition 3.3. A *canonical transformation* $(x, p, H) \mapsto (x', p', H')$ preserves the form of Hamilton's equations: $\dot{x}' = \partial H'/\partial p'$ and $\dot{p}' = -\partial H'/\partial x'$. Equivalently, it preserves the fundamental Poisson brackets, $\{x', x'\} = \{p', p'\} = 0$ and $\{x', p'\} = 1$.

Hamilton's equations follow from the phase-space form of Hamilton's principle,

$$\delta \int_{t_i}^{t_f} \mathrm{d}t \left[p\dot{x} - H(x, p, t)\right] = 0, \qquad \delta x(t_i) = \delta x(t_f) = 0. \tag{3.6}$$

(The handwritten notes also impose $\delta p = 0$ at the endpoints. That is Goldstein's convention; it is harmless but not needed.) Both sets of variables obey Hamilton's equations if the two integrands differ by a total time derivative:

$$p\dot{x} - H = p'\dot{x}' - H' + \frac{\mathrm{d}G}{\mathrm{d}t}. \tag{3.7}$$

For $G = G(x, x', t)$ the extra term contributes only the boundary term $[\partial_x G\,\delta x + \partial_{x'} G\,\delta x']_{t_i}^{t_f}$, which vanishes for variations that hold both x and x' fixed at the endpoints. That restriction is harmless, because Hamilton's equations come from the interior variations alone. Condition (3.7) is *sufficient*. The most general possibility also allows an overall constant factor (an "extended" canonical transformation), which we do not need.

Type-1 generating function. Take $G = G(x, x', t)$. Then $\frac{\mathrm{d}G}{\mathrm{d}t} = \partial_t G + \partial_x G\,\dot{x} + \partial_{x'} G\,\dot{x}'$, and (3.7) becomes

$$\Big(p - \frac{\partial G}{\partial x}\Big)\dot{x} - \Big(p' + \frac{\partial G}{\partial x'}\Big)\dot{x}' - H + H' - \frac{\partial G}{\partial t} = 0. \tag{3.8}$$

Since $\dot{x}$ and $\dot{x}'$ are independent, each bracket vanishes:

$$\boxed{p = \frac{\partial G}{\partial x}, \qquad p' = -\frac{\partial G}{\partial x'}, \qquad H' = H + \frac{\partial G}{\partial t}} \tag{3.9}$$

Generating functions of the other types, $G(x, p')$, $G(p, x')$ and $G(p, p')$, follow by Legendre transformation.

3.4 Hamilton–Jacobi theory: time evolution is a canonical transformation [hw. pp. 9–10]

Choose $G \equiv S(x, x', t)$ so that the new Hamiltonian vanishes, $H' = 0$. Then $\dot{x}' = \dot{p}' = 0$, so (x', p') are constants of motion. (If one only achieves $H' = f(t)$, the replacement $S \to S - \int f\,\mathrm{d}t$ removes it.) From (3.9):

1. The Hamilton–Jacobi equation:

$$H\Big(x, \frac{\partial S}{\partial x}, t\Big) + \frac{\partial S}{\partial t} = 0. \tag{3.10}$$

2. $p' = -\partial S/\partial x' = \text{const}$. Inverting gives $x = x(x', p', t)$, the solution of the equations of motion labelled by the constants (x', p').

3. Along that solution, $\frac{\mathrm{d}S}{\mathrm{d}t} = \partial_t S + \partial_x S\,\dot{x} = -H + p\dot{x} = L$, so

$$S = \int_{t_i}^{t} \mathrm{d}\tau\, L\big(x(\tau), \dot{x}(\tau), \tau\big) = S_{\mathrm{cl}}(x, t; x', t_i). \tag{3.11}$$

With $x' = x_i$, the relations (3.5) give $p' = -\partial S_{\mathrm{cl}}/\partial x_i = p_i$, so the constants are exactly the initial data. This is the key conclusion:

> The map $(x_i, p_i) \mapsto (x(t), p(t))$, time evolution itself, is a canonical transformation, and its type-1 generating function is Hamilton's principal function $S_{\mathrm{cl}}(x, t; x_i, t_i)$.

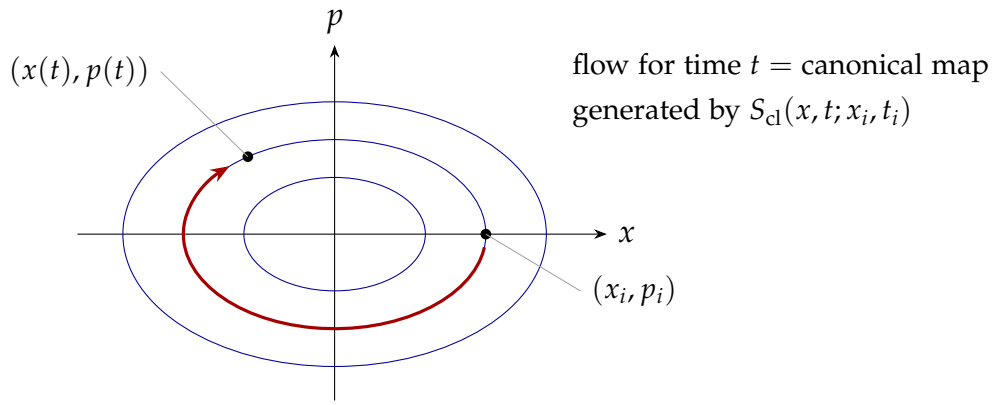


Figure 5: Phase space of the harmonic oscillator [hw. p. 3]. Time evolution carries (x_i, p_i) clockwise along an ellipse of constant energy. Hamilton–Jacobi theory identifies this flow as the canonical transformation generated by S_{cl}.

Remark 3.4 (The bridge to quantum mechanics). In quantum mechanics, time evolution is the *unitary* transformation $e^{-i\hat{H}t/\hbar}$. Dirac's insight was to ask which object in the quantum theory plays the role of the classical generating function. The natural candidate is the matrix element of the unitary map between position eigenstates, $\langle x, t | x', t' \rangle$. Section 4.2 makes this precise.

4 From Dirac's proposal to Feynman's path integral

This section follows the historical, heuristic route. Section 5 then derives the same result honestly from Schrödinger theory.

4.1 Position eigenstates and the propagator [hw. pp. 3, 12]

Let $\hat{x}(t) = e^{i\hat{H}t/\hbar}\,\hat{x}\,e^{-i\hat{H}t/\hbar}$ be the Heisenberg position operator, and define

$$|x,t\rangle \equiv e^{i\hat{H}t/\hbar}|x\rangle \quad \Longrightarrow \quad \hat{x}(t)|x,t\rangle = x|x,t\rangle. \tag{4.1}$$

At each fixed t these states form a complete set, $\int \mathrm{d}x\,|x,t\rangle\langle x,t| = \mathbb{1}$. The *propagator* is

$$K(f;i) \equiv \langle x_f,t_f|x_i,t_i\rangle = \langle x_f|e^{-i\hat{H}(t_f-t_i)/\hbar}|x_i\rangle \tag{4.2}$$

for a time-independent $\hat{H}$.

> **Caution: $|x,t\rangle$ is not a Schrödinger state**
> $|x,t\rangle = e^{+i\hat{H}t/\hbar}|x\rangle$ carries the opposite sign to Schrödinger evolution. It is the eigenstate of the *Heisenberg* operator $\hat{x}(t)$, not the state $|x\rangle$ evolved forward in time. The amplitude $\langle x_f,t_f|x_i,t_i\rangle$ nevertheless has the expected meaning: start localized at x_i at time t_i, evolve, and project onto x_f at time t_f.

Since $\int \mathrm{d}x\,|x\rangle\langle x| = \mathbb{1}$, the state $|x\rangle$ has dimension $(\text{length})^{-1/2}$, and therefore $[K] = (\text{length})^{-1}$. This fact is used in Section 4.4.

4.2 Dirac's proposal: a quantum generating function [hw. pp. 3, 10–11]

Section 3.4 showed that time evolution is the canonical transformation generated by S_{cl}. Dirac asked whether $\langle x,t|x',t'\rangle$ plays the role of a generating function. Write it, without loss of generality, as

$$\langle x,t|x',t'\rangle \equiv e^{iG(x,x')/\hbar}, \qquad t > t'. \tag{4.3}$$

Using $\langle x,t|\hat{p}(t) = -i\hbar\,\partial_x\langle x,t|$ and $\hat{p}(t')|x',t'\rangle = +i\hbar\,\partial_{x'}|x',t'\rangle$ (in the sense of matrix elements),

$$\langle x,t|\hat{p}(t)|x',t'\rangle = \frac{\partial G}{\partial x}\,\langle x,t|x',t'\rangle, \qquad \langle x,t|\hat{p}(t')|x',t'\rangle = -\frac{\partial G}{\partial x'}\,\langle x,t|x',t'\rangle. \tag{4.4}$$

These are the exact quantum analogues of the classical relations $p = \partial G/\partial x$ and $p' = -\partial G/\partial x'$ in (3.9). As operator statements they read $\hat{p}(t) = \frac{\partial G}{\partial x}(\hat{x}(t),\hat{x}(t'))$ and $\hat{p}(t') = -\frac{\partial G}{\partial x'}(\hat{x}(t),\hat{x}(t'))$, provided every $\hat{x}(t)$ stands to the *left* of every $\hat{x}(t')$ (Dirac's "well-ordered" functions). Only then does $\langle x,t|f(\hat{x}(t))g(\hat{x}(t'))|x',t'\rangle = f(x)g(x')\langle x,t|x',t'\rangle$ hold.

Since classically the generator of time evolution is S_{cl}, Dirac proposed

$$\langle x_f, t_f | x_i, t_i \rangle \text{ “corresponds to” } e^{iS_{\mathrm{cl}}(x_f,t_f;x_i,t_i)/\hbar}, \qquad \text{i.e.} \quad G = S_{\mathrm{cl}} + O(\hbar). \tag{4.5}$$

Example 4.1 (Free particle: what G really is)**.** The exact free propagator (derived in Section 9.4) is $K = \sqrt{\frac{m}{2\pi i\hbar T}}\, e^{im(x-x')^2/2\hbar T}$, so

$$G = \underbrace{\frac{m(x-x')^2}{2T}}_{S_{\mathrm{cl}}} - \frac{i\hbar}{2} \ln \frac{m}{2\pi i\hbar T}. \tag{4.6}$$

G is complex and differs from S_{cl} at order $\hbar$. The correction is the logarithm of the prefactor. The relations (4.4) still hold: $\partial_x G = -\partial_{x'} G = m(x-x')/T$ reproduces $\hat{p} = m\big(\hat{x}(t) - \hat{x}(t')\big)/T$, which follows from $\hat{x}(t) = \hat{x}(t') + \hat{p}\, T/m$. Here ordering does not matter because everything is linear.

4.3 Why Dirac's formula is only semiclassical [hw. p. 3]

The handwritten argument notes that the left-hand side of (4.5) is independent of any path, whereas $e^{iS[x]/\hbar}$ depends on which path is chosen. That is correct for the functional $S[x]$. With S_{cl} the right-hand side is also path-independent, and what actually fails is the following:

1. The **prefactor** is missing. Even for the free particle, $e^{iS_{\mathrm{cl}}/\hbar}$ alone is wrong; the factor $\sqrt{m/2\pi i\hbar T}$ is needed.
2. **Several classical paths** may connect the same endpoints, and all of them contribute. For the oscillator at $\omega T = \pi$ there are infinitely many.
3. The formula is exact only for **infinitesimal times**, or for quadratic Lagrangians up to the prefactor.

Dirac's formula is therefore the leading semiclassical (WKB) approximation. To go beyond it we need a way to compose short times into finite ones.

4.4 Composition: time slicing [hw. pp. 4–5]

The action is additive along a path. Splitting $[t_i, t_f]$ into N slices,

$$S[x] = \sum_{n=0}^{N-1} S_{n,n+1}, \qquad S_{n,n+1} = \int_{t_n}^{t_{n+1}} L\, \mathrm{d}t. \tag{4.7}$$

This suggests two ways to build the finite-time amplitude from short-time pieces.

Way 1 (classical intuition). Pick intermediate points $\tilde{x}_1, \ldots, \tilde{x}_{N-1}$ on one path and multiply:

$$K(f;i) \stackrel{?}{=} \langle x_f, t_f | \tilde{x}_{N-1}, t_{N-1} \rangle \cdots \langle \tilde{x}_1, t_1 | x_i, t_i \rangle. \tag{4.8}$$

Way 2 (quantum mechanics). Insert the completeness relation at each intermediate time:

$$K(f;i) = \int \mathrm{d}x_1 \cdots \mathrm{d}x_{N-1} \, \langle x_f, t_f | x_{N-1}, t_{N-1} \rangle \cdots \langle x_1, t_1 | x_i, t_i \rangle. \tag{4.9}$$

Way 1 cannot be right. It singles out one path, and it is not even dimensionally consistent: each factor has dimension 1/length, so the product has dimension $1/\text{length}^N$, while K has dimension 1/length. In Way 2 the $N-1$ integrations supply exactly the missing length^{N-1}. *Integrating over the intermediate positions is forced by quantum mechanics, and it is precisely this integration that turns "one path" into "all paths".*

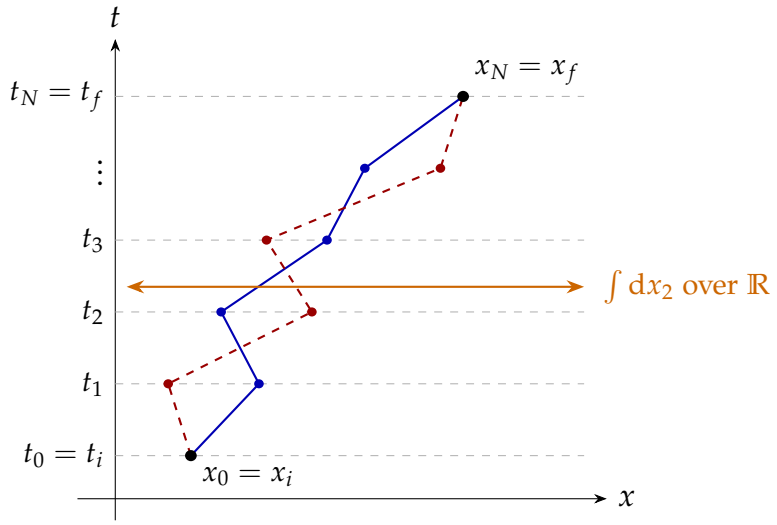


Figure 6: Time slicing. Each choice of intermediate positions $(x_1, \ldots, x_{N-1})$ defines a polygonal path; two are shown. Integrating each x_n over $\mathbb{R}$ sums over all polygonal paths, and as $N \to \infty$ these approximate all continuous paths.

4.5 Feynman's prescription [hw. pp. 5–6]

Feynman combined the two ingredients. Dirac's formula is used where it is reliable, for infinitesimal times, and Way 2 is used to compose:

$$\langle x_{n+1}, t_n + \varepsilon | x_n, t_n \rangle = A \, \exp\left[\frac{i}{\hbar} \varepsilon \, L\left(x_n, \frac{x_{n+1} - x_n}{\varepsilon}\right)\right] \big(1 + O(\varepsilon)\big), \tag{4.10}$$

where εL approximates S_{cl} for the short segment. For smooth potentials it does not matter at which point of the segment V is evaluated. For velocity-dependent couplings (a particle in a magnetic field) it does matter; see Section 5.2. Substituting (4.10) into (4.9):

Feynman's path integral

$$K(f;i) = \lim_{\substack{N\to\infty \\ N\varepsilon = T}} A^N \int \prod_{n=1}^{N-1} \mathrm{d}x_n \, \exp\Big[\frac{i}{\hbar}\sum_{n=0}^{N-1} \varepsilon\, L\Big(x_n, \frac{x_{n+1}-x_n}{\varepsilon}\Big)\Big] \equiv \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x \; e^{iS[x]/\hbar}. \tag{4.11}$$

The constant A is fixed by matching to Schrödinger theory: $A = \sqrt{m/2\pi i\hbar\varepsilon}$ (Section 5.3). In the limit, $\sum_n \varepsilon L_n \to \int L\,\mathrm{d}t = S[x]$ is just a Riemann sum.

Remark 4.2 ([hw. p. 6]). In quantum mechanics the *entire functional form* of $S[x]$ matters, not only its value at the extremum. Every path contributes, with weight of modulus one.

4.6 The classical limit: stationary phase in path space [hw. pp. 6–7]

The integral (4.11) has the structure of Section 2.5, with $\lambda \leftrightarrow 1/\hbar$ and the integration variable now the whole path. Expand about the classical path, $x = x_{\mathrm{cl}} + \eta$:

$$S[x_{\mathrm{cl}}+\eta] = S_{\mathrm{cl}} + \underbrace{\int \mathrm{d}t \, \frac{\delta S}{\delta x(t)}\Big|_{x_{\mathrm{cl}}} \eta(t)}_{=0} + \tfrac{1}{2}\,\delta^2 S[\eta] + \cdots \tag{4.12}$$

- **Far from** x_{cl}, a small change of path changes the phase $S/\hbar$ by a large amount when $S \gg \hbar$. The factor $e^{iS/\hbar}$ sweeps round the unit circle (its real and imaginary parts oscillate between -1 and $+1$), and neighbouring paths cancel.
- **Near** x_{cl}, $\delta S = 0$, so the phase changes only at second order. Paths within a "tube" where $|\delta^2 S| \lesssim \hbar$ add coherently. Estimating $\delta^2 S \sim m w^2/T$ for a deviation of size w gives a tube width
$$w \sim \sqrt{\hbar T/m}. \tag{4.13}$$
- As $\hbar \to 0$ the tube collapses onto x_{cl}, and the condition $\delta S = 0$ is the Euler–Lagrange equation. Classical mechanics is the stationary-phase limit of the sum over histories.

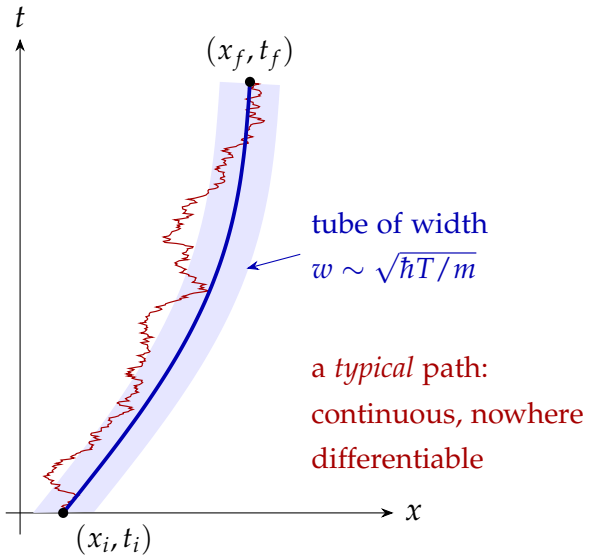


Figure 7: The paths that survive the cancellation lie in a tube (shaded) around x_{cl} (thick curve). The classical path itself has measure zero: it is the tube, not the line, that produces the prefactor. A typical contributing path (thin jagged curve) has $|x_{n+1} - x_n| \sim \sqrt{\hbar\varepsilon/m}$.

Caution: typical paths are not differentiable

From the short-time kernel and (2.15), the second moment of a single step is

$$\begin{aligned} \big\langle (x_{n+1} - x_n)^2 \big\rangle &= \frac{i\hbar\varepsilon}{m}, \\ \text{so} \qquad |x_{n+1} - x_n| \sim \sqrt{\hbar\varepsilon/m}, &\qquad \frac{|x_{n+1} - x_n|}{\varepsilon} \sim \varepsilon^{-1/2} \to \infty. \end{aligned} \tag{4.14}$$

Paths that dominate the integral are continuous but have no velocity. Expressions such as $\int \dot{x}^2 \, \mathrm{d}t$ are defined only through their discretization. This is the root of the operator-ordering subtleties in Sections 5.2 and 7.7.

5 Derivation from Schrödinger theory

Section 4 *postulated* the short-time kernel (4.10). Here we derive it from canonical quantum mechanics, and with it the full path integral. The result is first a path integral over phase space. The configuration-space form of Feynman follows for a large class of Hamiltonians, together with a precise account of when and how it can fail.

5.1 Time slicing the evolution operator [hw. p. 12]

With $T = N\varepsilon$, write $e^{-i\hat{H}T/\hbar} = \left(e^{-i\hat{H}\varepsilon/\hbar}\right)^N$ and insert $\int \mathrm{d}x_n \, |x_n\rangle\langle x_n| = \mathbb{1}$ between consecutive factors:

$$K(f;i) = \lim_{N\to\infty} \int \prod_{n=1}^{N-1} \mathrm{d}x_n \prod_{n=0}^{N-1} \langle x_{n+1}|e^{-i\hat{H}\varepsilon/\hbar}|x_n\rangle, \qquad x_0 = x_i,\ x_N = x_f. \tag{5.1}$$

Everything now rests on the short-time matrix element.

5.2 The phase-space path integral [hw. p. 13]

Insert a momentum resolution of the identity in each short-time factor:

$$\langle x_{n+1}|e^{-i\hat{H}\varepsilon/\hbar}|x_n\rangle = \int \mathrm{d}p_n \, \langle x_{n+1}|p_n\rangle\langle p_n|e^{-i\hat{H}\varepsilon/\hbar}|x_n\rangle. \tag{5.2}$$

Define the c-number function

$$\boxed{H(p,x) \equiv \frac{\langle p|\hat{H}|x\rangle}{\langle p|x\rangle}} \tag{5.3}$$

Then $\langle p|e^{-i\hat{H}\varepsilon/\hbar}|x\rangle = \langle p|x\rangle\left[1 - \frac{i\varepsilon}{\hbar}H(p,x) + O(\varepsilon^2)\right] = \langle p|x\rangle\, e^{-i\varepsilon H(p,x)/\hbar}\left[1 + O(\varepsilon^2)\right]$. The re-exponentiation error is $O(\varepsilon^2)$ per slice, hence $O(\varepsilon)$ in total; the Trotter product formula makes this rigorous for well-behaved $\hat{H}$. With $\langle x_{n+1}|p_n\rangle\langle p_n|x_n\rangle = e^{ip_n(x_{n+1}-x_n)/\hbar}/2\pi\hbar$:

Phase-space (Hamiltonian) path integral

$$K(f;i) = \lim_{N\to\infty} \int \prod_{n=1}^{N-1} \mathrm{d}x_n \prod_{n=0}^{N-1} \frac{\mathrm{d}p_n}{2\pi\hbar} \, \exp\left\{\frac{i\varepsilon}{\hbar}\sum_{n=0}^{N-1}\left[p_n\frac{x_{n+1}-x_n}{\varepsilon} - H(p_n,x_n)\right]\right\} \tag{5.4}$$

$$\equiv \int \mathcal{D}x\,\mathcal{D}p \, \exp\left\{\frac{i}{\hbar}\int_{t_i}^{t_f} \mathrm{d}t\,\left[p\dot{x} - H(p,x)\right]\right\}. \tag{5.5}$$

Three features deserve comment.

1. **Counting.** There are $N-1$ position integrals but N momentum integrals. The boundary data fix x at both ends and leave p free. The measure $\mathcal{D}x\,\mathcal{D}p$ is therefore *not* the symmetric Liouville measure on phase space.

2. **Ordering.** Definition (5.3) is exact, but it is an ordering prescription. $H(p,x)$ equals the classical function obtained by writing $\hat{H}$ with every $\hat{p}$ to the *left* of every $\hat{x}$. For $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ there is no ambiguity and $H = p^2/2m + V(x)$. For mixed terms there is. For example, $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) = \hat{p}\hat{x} + \frac{i\hbar}{2}$ gives $H(p,x) = px + \frac{i\hbar}{2}$, which is complex. Evaluating H at the midpoint $\bar{x}_n = \frac{1}{2}(x_n + x_{n+1})$ instead corresponds to *Weyl* ordering, in which the symbol of $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ is exactly xp. Ordering ambiguities are $O(\hbar)$. Because typical paths have $\Delta x \sim \sqrt{\varepsilon}$ (4.14), they survive the continuum limit whenever H couples p to x, as for a particle in a magnetic field.

3. **Generality.** No assumption about the form of $\hat{H}$ has been made. This is the most general form of the path integral, and the starting point for constrained systems.

5.3 Case 1: Hamiltonians quadratic in momentum [hw. pp. 14–15]

For $\hat{H} = \hat{p}^2/2m + V(\hat{x})$, complete the square in each slice, with $\dot{x}_n \equiv (x_{n+1} - x_n)/\varepsilon$:

$$p_n\dot{x}_n - \frac{p_n^2}{2m} = -\frac{1}{2m}\big(p_n - m\dot{x}_n\big)^2 + \frac{1}{2}m\dot{x}_n^2. \tag{5.6}$$

Shift $q_n = p_n - m\dot{x}_n$ in each p_n integral (unit Jacobian at fixed x's). Each momentum integral then factorizes and is a Fresnel integral (2.14):

$$\int \frac{\mathrm{d}q_n}{2\pi\hbar}\, e^{-i\varepsilon q_n^2/2m\hbar} = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m\hbar}{i\varepsilon}} = \sqrt{\frac{m}{2\pi i\hbar\varepsilon}} \equiv A. \tag{5.7}$$

There are N such factors, so the "constant absorbed into the measure" [hw. p. 15] is exactly A^N:

Configuration-space (Lagrangian) path integral

$$K(f;i) = \lim_{N\to\infty}\Big(\frac{m}{2\pi i\hbar\varepsilon}\Big)^{N/2}\int\prod_{n=1}^{N-1}\mathrm{d}x_n\ \exp\Big\{\frac{i}{\hbar}\sum_{n=0}^{N-1}\varepsilon\Big[\frac{m}{2}\Big(\frac{x_{n+1}-x_n}{\varepsilon}\Big)^2 - V(x_n)\Big]\Big\}. \tag{5.8}$$

This is Feynman's formula (4.11), now *derived*. As a by-product it fixes the normalization $A = \sqrt{m/2\pi i\hbar\varepsilon}$, which on [hw. p. 6] was to be "fixed by matching".

Remark 5.1. The momentum integral is a stationary-phase integral in p. Its stationary point $p = m\dot{x}$ is the Legendre transform, and the evaluation is *exact* because the exponent is quadratic in p. Case 2 generalizes exactly this observation.

5.4 Case 2: general Hamiltonians [hw. pp. 15–16]

Let $\ell(p) \equiv p\dot{x} - H(p,x,t)$. Its stationary point p_0 satisfies

$$\partial_p \ell\big|_{p_0} = 0 \quad \Longleftrightarrow \quad \dot{x} = \frac{\partial H}{\partial p}\Big|_{p_0}, \tag{5.9}$$

which is half of Hamilton's equations. Inverting (possible when $\partial_p^2 H \neq 0$) gives $p_0 = p_0(x,\dot{x},t)$, and $\ell(p_0) = L(x,\dot{x},t)$ is the Legendre transform. Taylor expanding around p_0, where the linear term vanishes by stationarity,

$$\begin{aligned} p\dot{x} - H &= L(x,\dot{x},t) + \sum_{n\geq 2} K_n(x,\dot{x},t)\,(p-p_0)^n, \\ K_n &= \frac{1}{n!}\partial_p^n \ell\Big|_{p_0} = -\frac{1}{n!}\partial_p^n H\Big|_{p_0} \qquad (n \geq 2). \end{aligned} \tag{5.10}$$

The path integral factorizes into a configuration-space part and a momentum part:

$$K(f;i) = \int \mathcal{D}x\; e^{iS[x]/\hbar}\; \mathcal{M}[x], \qquad \mathcal{M}[x] = \int \mathcal{D}p\; \exp\Big\{\frac{i}{\hbar}\int \mathrm{d}t \sum_{n\geq 2} K_n\,(p-p_0)^n\Big\}. \tag{5.11}$$

Rescale $p - p_0 = \sqrt{\hbar}\,q$. This gives one constant factor $\hbar^{1/2}$ *per time slice*, absorbed into the normalization. Writing $\langle \cdots \rangle \equiv \int \mathrm{d}t\,(\cdots)$,

$$\begin{aligned} \mathcal{M} &\propto \int \mathcal{D}q\; \exp\Big\{ i\langle K_2 q^2\rangle + i\hbar^{1/2}\langle K_3 q^3\rangle + i\hbar\langle K_4 q^4\rangle + \cdots \Big\} \\ &= \int \mathcal{D}q\; e^{i\langle K_2 q^2\rangle}\Big[1 + i\hbar^{1/2}\langle K_3 q^3\rangle + \hbar\Big(i\langle K_4 q^4\rangle - \tfrac{1}{2}\langle K_3 q^3\rangle^2\Big) + O(\hbar^{3/2})\Big]. \end{aligned} \tag{5.12}$$

(The handwritten expansion omitted the factors of i and had the wrong sign on the $\langle K_3 q^3\rangle^2$ term. Neither affects the parity argument below, but both affect M_1.)

Lemma 5.2 (Only integer powers of $\hbar$ survive)**.** *A general term in the expansion is a product $\prod_{j=1}^{k}\langle K_{m_j} q^{m_j}\rangle$ with $m_j \geq 3$. It carries the power $\hbar^{\sum_j (m_j-2)/2}$ and contains $\sum_j m_j = 2k + \sum_j (m_j - 2)$ factors of q. Hence a half-integer power of $\hbar$ comes with an odd number of q's, and its integral against the even Gaussian weight $e^{i\langle K_2 q^2\rangle}$ vanishes.*

Therefore $\mathcal{M} = M_0 + \hbar M_1 + \hbar^2 M_2 + \cdots$, and

$$\boxed{K(f;i) = \int \mathcal{D}x\; \mathcal{M}[x]\; e^{iS[x]/\hbar}} \tag{5.13}$$

The leading factor M_0 is not a constant. In discretized form,

$$M_0 = \prod_n \int \mathrm{d}q_n\, e^{i\varepsilon K_2^{(n)} q_n^2} = \prod_n \sqrt{\frac{i\pi}{\varepsilon K_2^{(n)}}}, \qquad K_2 = -\tfrac{1}{2}\partial_p^2 H. \tag{5.14}$$

So

$$M_0 \;\propto\; \prod_t \Big[\partial_p^2 H\big(p_0(x,\dot{x},t),x,t\big)\Big]^{-1/2}, \tag{5.15}$$

a *local* (time-by-time) modification of the measure. It is path-dependent unless $\partial_p^2 H$ is constant.

- $H = p^2/2m$: $\partial_p^2 H = 1/m$ is constant, and one recovers A^N (Case 1).
- Position-dependent mass, $H = p^2/2m(x)$: the measure becomes

$$\prod_n \mathrm{d}x_n \sqrt{\frac{m(x_n)}{2\pi i\hbar\varepsilon}}. \tag{5.16}$$

- A particle on a curved space, $H = \frac{1}{2}g^{ij}(x)p_i p_j + V$: the measure becomes

$$\prod_n \sqrt{\det g_{ij}(x_n)}\, \mathrm{d}^d x_n, \tag{5.17}$$

  the Riemannian volume element. The naive flat measure is wrong.

Caution: the expansion is formal

Formally $\prod_t f(t) = \exp\left[\delta(0)\int \mathrm{d}t\, \ln f(t)\right]$, with $\delta(0) \leftrightarrow 1/\varepsilon$. Moreover the q-integral is *ultralocal*: $\langle q(t)q(t')\rangle \propto \delta(t-t')/K_2$. The contractions in M_1 therefore produce further factors of $\delta(0)$. These Lee–Yang $\delta(0)$ terms are meaningful only in the discretized theory, and in perturbation theory they typically cancel against divergences elsewhere. The continuum notation hides them.

5.5 Conclusion: two stationary-phase steps [hw. p. 17]

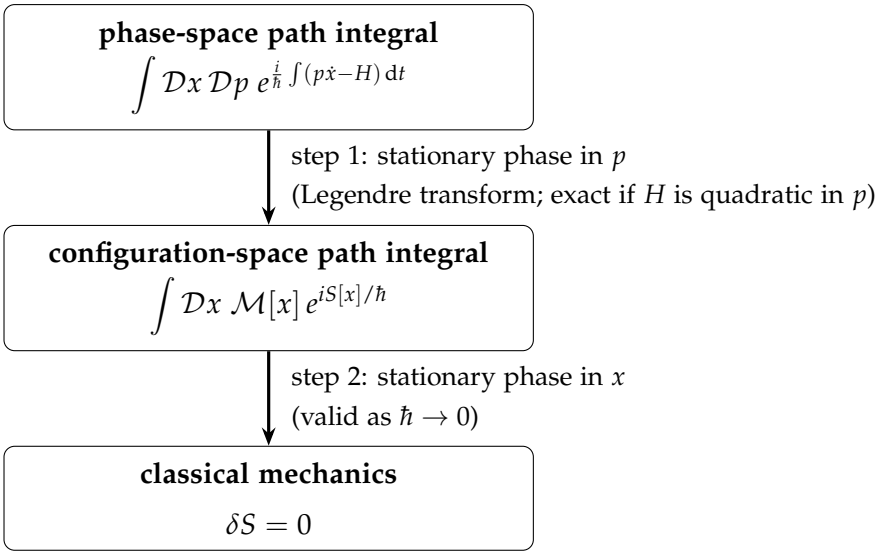


Figure 8: The passage from Schrödinger theory to classical mechanics is two stationary-phase evaluations. Fluctuations around each saddle are the two sources of quantum corrections.

Viewed as an expansion around the stationary point (a semiclassical expansion), the general result (5.13) has a classical leading term and two sources of quantum corrections:

1. **Off-classical paths in configuration space.** Fluctuations of $x(t)$ around $x_{\rm cl}$ produce the prefactor (Van Vleck determinant) and higher loop corrections.
2. **Local modification of the measure.** This is $\mathcal{M}[x]$, from fluctuations of p around p_0, including the operator-ordering ambiguity hidden in the definition of $H(p,x)$.

> **Caution: when the Legendre transform does not exist**
>
> Step 1 requires $\det\left(\partial^2 H/\partial p_i \partial p_j\right) \neq 0$. It fails for reparametrization-invariant and gauge systems: the relativistic particle, whose Hamiltonian is a constraint ($p^2 + m^2 \approx 0$ in mostly-plus signature), gauge theories, and gravity. There the phase-space path integral (5.4), supplemented by gauge fixing, is the fundamental object. This is the starting point of the relativistic part of the course.

5.6 Consistency check: recovering the Schrödinger equation★

Apply one short-time kernel (5.8) to a wavefunction:

$$\psi(x, t+\varepsilon) = A \int \mathrm{d}x' \, \exp\Big[\frac{im(x-x')^2}{2\hbar\varepsilon} - \frac{i\varepsilon}{\hbar} V(x)\Big] \psi(x', t). \tag{5.18}$$

(The kernel (5.8) evaluates V at the earlier point x'. Using the later point x changes the result only at $O(\varepsilon^2)$, because $V(x) - V(x') = O(\xi)$ and ξ has zero mean.) Let $\xi = x - x'$. The Gaussian restricts $\xi \sim \sqrt{\hbar\varepsilon/m}$, so expand $\psi(x-\xi, t) = \psi - \xi\psi' + \frac{1}{2}\xi^2\psi'' + \cdots$ and $e^{-i\varepsilon V/\hbar} = 1 - i\varepsilon V/\hbar + \cdots$. With $a = m/2\hbar\varepsilon$ in (2.14)–(2.15),

$$A\int \mathrm{d}\xi \, e^{ia\xi^2} = 1, \qquad A\int \mathrm{d}\xi \, \xi \, e^{ia\xi^2} = 0, \qquad A\int \mathrm{d}\xi \, \xi^2 e^{ia\xi^2} = \frac{i\hbar\varepsilon}{m}. \tag{5.19}$$

Hence $\psi(x, t+\varepsilon) = \psi + \varepsilon\big[\frac{i\hbar}{2m}\psi'' - \frac{i}{\hbar}V\psi\big] + O(\varepsilon^2)$. Letting $\varepsilon \to 0$ and multiplying by $i\hbar$,

$$i\hbar\, \partial_t \psi = -\frac{\hbar^2}{2m}\partial_x^2 \psi + V(x)\psi. \tag{5.20}$$

The path integral and canonical quantum mechanics are the same theory. The fact that the zeroth moment is exactly 1 is an independent check of the value of A.

6 The propagator as a Green's function

6.1 Evolution of wavefunctions [hw. p. 18]

The Schrödinger wavefunction is $\psi(x,t) = \langle x|\psi(t)\rangle$. Inserting a complete set at t_i,

$$\psi(x_f,t_f) = \langle x_f|e^{-i\hat{H}(t_f-t_i)/\hbar}|\psi(t_i)\rangle = \int \mathrm{d}x_i\, K(f;i)\,\psi(x_i,t_i). \tag{6.1}$$

The propagator is the amplitude for each point of the initial wavefunction to feed the value at (x_f,t_f), in the spirit of Huygens' principle (Fig. 9). Inserting a complete set at an intermediate time gives the *composition law*

$$K(f;i) = \int \mathrm{d}x\, K(x_f,t_f;x,t)\,K(x,t;x_i,t_i), \qquad t_i < t < t_f, \tag{6.2}$$

which is the property that made time slicing possible.

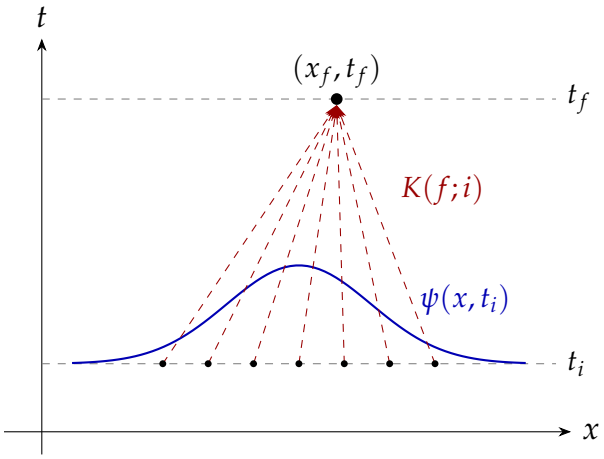


Figure 9: Every point x_i of the initial wavefunction contributes to $\psi(x_f,t_f)$ with amplitude $K(f;i)$, and (6.1) adds the contributions.

6.2 The propagator solves the Schrödinger equation [hw. p. 18]

Proposition 6.1.

$$i\hbar\,\partial_{t_f}K(f;i) = H\big(-i\hbar\partial_{x_f},x_f\big)K(f;i), \qquad \lim_{t_f\to t_i} K(f;i) = \delta(x_f-x_i). \tag{6.3}$$

Proof. Directly: $i\hbar\,\partial_{t_f}\langle x_f|e^{-i\hat{H}T/\hbar}|x_i\rangle = \langle x_f|\hat{H}\,e^{-i\hat{H}T/\hbar}|x_i\rangle$, and in the position representation $\hat{H}$ acts on x_f as $H(-i\hbar\partial_{x_f},x_f)$. At $T=0$ the matrix element is $\langle x_f|x_i\rangle = \delta(x_f-x_i)$.

Via wavefunctions (the handwritten hint): $\psi(f)$ obeys the Schrödinger equation for every initial $\psi(i)$. Inserting (6.1) gives $\int \mathrm{d}x_i\,[(i\hbar\partial_{t_f}-H)K]\,\psi(x_i,t_i) = 0$ for all $\psi(i)$, so the bracket vanishes. □

Remark 6.2. For Hamiltonians that mix x and p, the differential operator $H(-i\hbar\partial_x, x)$ inherits the ordering ambiguity of Section 5.2. It must be understood with the same ordering as $\hat{H}$. For $\hat{p}^2/2m + V(\hat{x})$ there is no issue.

6.3 The retarded propagator [hw. pp. 19–20]

Define $K_r(f;i) \equiv \theta(t_f - t_i)\, K(f;i)$. Since $\theta' = \delta$ (Section 2.4),

$$\begin{aligned} i\hbar\,\partial_{t_f} K_r &= i\hbar\,\delta(t_f - t_i)\, K(f;i) + \theta(t_f - t_i)\, i\hbar\,\partial_{t_f} K(f;i) \\ &= i\hbar\,\delta(t_f - t_i)\, K(x_f, t_i; x_i, t_i) + H\big(-i\hbar\partial_{x_f}, x_f\big) K_r(f;i), \end{aligned} \tag{6.4}$$

where (2.9) was used in the first term and (6.3) in the second. With $K(x_f, t_i; x_i, t_i) = \delta(x_f - x_i)$:

K_r **is the Green's function of the Schrödinger operator**
$$\Big[i\hbar\,\partial_t - H\big(-i\hbar\partial_x, x\big)\Big] K_r(x, t; x_i, t_i) = i\hbar\,\delta(t - t_i)\,\delta(x - x_i). \tag{6.5}$$

Consequently the inhomogeneous equation $(i\hbar\partial_t - H)\psi = j$ is solved by $\psi(x,t) = \frac{1}{i\hbar}\int \mathrm{d}t'\,\mathrm{d}x'\, K_r(x,t;x',t')\, j(x',t')$.

Remark 6.3 (The margin question of hw. p. 20). *Why does the limit $t_f \to t_i$ apply only to the K multiplying the delta function, and not to the whole equation?* Because $\delta(t_f - t_i)\, g(t_f) = \delta(t_f - t_i)\, g(t_i)$ (2.9): a delta function evaluates only its own coefficient at $t_f = t_i$. The second term, HK_r, has no delta function in front of it, so nothing in it is evaluated.

A subtlety: for fixed $x_f \neq x_i$, $K(x_f, t; x_i, t_i)$ is *not* continuous as $t \to t_i$. The free propagator oscillates infinitely fast there. The clean statement is at the operator level. With $\hat{U}(T) = e^{-i\hat{H}T/\hbar}$ and $\hat{U}_r(T) = \theta(T)\hat{U}(T)$,

$$i\hbar\,\frac{\mathrm{d}\hat{U}_r}{\mathrm{d}T} = i\hbar\,\delta(T)\,\hat{U}(T) + \theta(T)\,\hat{H}\,\hat{U}(T) = i\hbar\,\delta(T)\,\mathbb{1} + \hat{H}\,\hat{U}_r(T), \tag{6.6}$$

since $\hat{U}(T) \to \mathbb{1}$ as $T \to 0$. Sandwiching between $\langle x_f|$ and $|x_i\rangle$ gives (6.5).

6.4 Other Green's functions and the energy representation★

Equation (6.5) does not determine its solution uniquely; boundary conditions do. The *advanced* propagator $K_a(f;i) \equiv -\theta(t_i - t_f)K(f;i)$ satisfies the same equation (check: $\partial_T[-\theta(-T)] = \delta(T)$). Their difference $K_r - K_a = K$ solves the homogeneous equation. The retarded choice encodes causality: a source acts only on the future.

Fourier transforming in the elapsed time reveals the structure, with the regulator $e^{-\eta T/\hbar}$ supplying convergence:

$$\hat{G}_r(E) \equiv \int_{-\infty}^{\infty} \mathrm{d}T\, e^{iET/\hbar}\, \theta(T)\, e^{-i\hat{H}T/\hbar} = \frac{i\hbar}{E - \hat{H} + i\eta} = \sum_n \frac{i\hbar\, |n\rangle\langle n|}{E - E_n + i\eta}, \tag{6.7}$$

and similarly $\hat{G}_a(E) = i\hbar/(E - \hat{H} - i\eta)$. The poles of $\hat{G}_r$ sit just *below* the real energy axis, at $E_n - i\eta$; those of $\hat{G}_a$ sit just above (Fig. 10). Causality ($\theta(T)$) is equivalent to analyticity of $\hat{G}_r(E)$ in the upper half-plane, because there $e^{iET/\hbar}$ decays for $T > 0$. This is the same $+i\eta$ as in the Feynman propagator of quantum field theory.

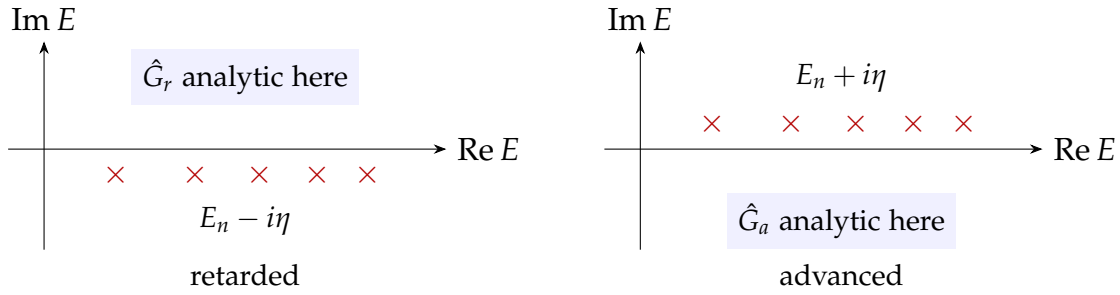


Figure 10: Poles of the retarded and advanced Green's functions in the complex energy plane.

Remark 6.4 (Non-relativistic retarded = Feynman). In non-relativistic quantum mechanics the spectrum is bounded below and there are no negative-frequency (antiparticle) modes, so the time-ordered and retarded propagators coincide. In second-quantized language $\hat{\psi}|0\rangle = 0$, so $\langle 0|\mathcal{T}\,\hat{\psi}(x,t)\hat{\psi}^\dagger(x',t')|0\rangle = \theta(t-t')K(x,t;x',t')$. The two propagators differ only in relativistic theory, where the negative-frequency pole must propagate backwards in time and the Feynman prescription places it above the real axis.

7 Insertions, time ordering and the Schwinger–Dyson equation

So far $\int \mathcal{D}x\, e^{iS/\hbar} = \langle x_f, t_f | x_i, t_i \rangle$. This section asks what happens when classical functions of the path are inserted into the integral:

$$\int \mathcal{D}x\; e^{iS[x]/\hbar}\,(\cdots) \quad \longleftrightarrow \quad \langle x_f, t_f | (\cdots) | x_i, t_i \rangle, \tag{7.1}$$

with classical variables built from x and $\dot{x}$ on the left and quantum operators on the right. In what follows $\langle f| \equiv \langle x_f, t_f|$ and $|i\rangle \equiv |x_i, t_i\rangle$.

7.1 Cutting open and sewing [hw. pp. 20–21]

The composition law (6.2), written as path integrals, says that the integral over all paths from i to f can be *cut open* at any intermediate time t and *sewn* back by integrating over the crossing point:

$$\int_i^f \mathcal{D}x\, e^{\frac{i}{\hbar}S_{if}[x]} = \int \mathrm{d}x \underbrace{\int_{(x,t)}^{f} \mathcal{D}x\, e^{\frac{i}{\hbar}S_{(x,t)f}[x]}}_{\langle f|x,t\rangle} \underbrace{\int_{i}^{(x,t)} \mathcal{D}x\, e^{\frac{i}{\hbar}S_{i(x,t)}[x]}}_{\langle x,t|i\rangle}. \tag{7.2}$$

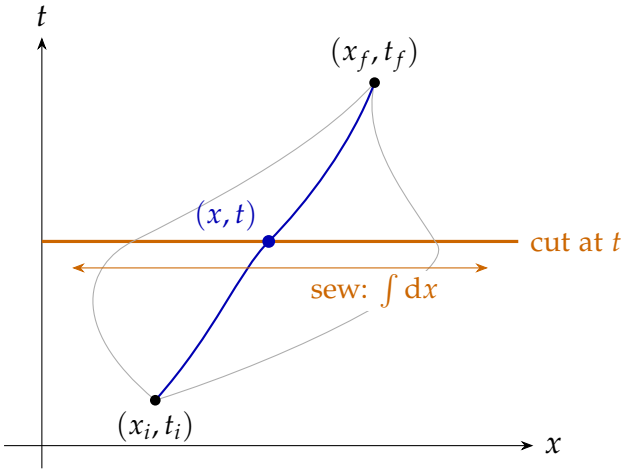


Figure 11: Cutting the path integral open at an intermediate time and sewing it back with $\int \mathrm{d}x$. Each path crosses the cut exactly once, at some x.

7.2 One insertion [hw. p. 21]

Inserting the classical function $x(t)$, cutting at t, and noting that the inserted value is the crossing point x:

$$\int \mathcal{D}x\, e^{iS/\hbar}\, x(t) = \int \mathrm{d}x\, \langle f|x,t\rangle\, x\, \langle x,t|i\rangle = \int \mathrm{d}x\, \langle f|\hat{x}(t)|x,t\rangle\langle x,t|i\rangle = \langle f|\hat{x}(t)|i\rangle. \tag{7.3}$$

Here $\hat{x}(t)|x,t\rangle = x|x,t\rangle$ was used, followed by completeness.

7.3 Two insertions: time ordering emerges [hw. pp. 21–22, 32]

Classically $x(t)x(t') = x(t')x(t)$, but operators do not commute, so which order should appear? Cut at both times. If $t > t'$, the cut at t lies above the cut at t', and sewing gives $\langle f|\hat{x}(t)\hat{x}(t')|i\rangle$. If $t' > t$ the order reverses. In both cases the later time stands to the left:

$$\begin{aligned}\int \mathcal{D}x\, e^{iS/\hbar}\, x(t)\, x(t') &= \langle f|\big[\theta(t-t')\hat{x}(t)\hat{x}(t') + \theta(t'-t)\hat{x}(t')\hat{x}(t)\big]|i\rangle \\ &\equiv \langle f|\mathcal{T}\big[\hat{x}(t)\hat{x}(t')\big]|i\rangle.\end{aligned} \tag{7.4}$$

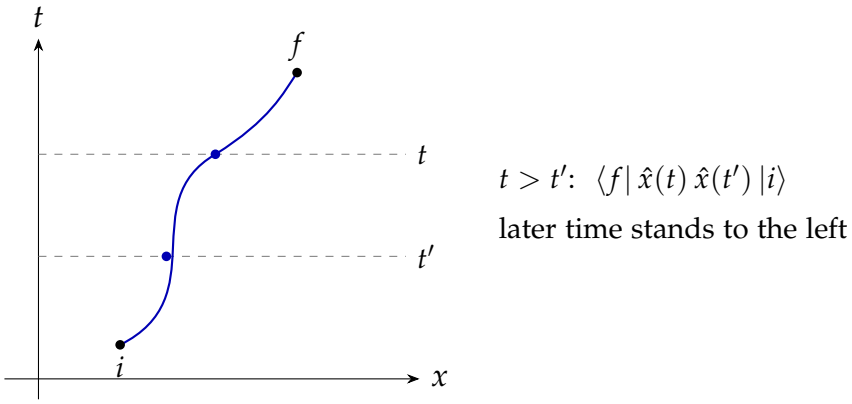


Figure 12: With two insertions, sewing the cuts in temporal order produces a time-ordered product.

General insertions

$$\int \mathcal{D}x\, e^{iS/\hbar}\, x(t_1)\cdots x(t_n) = \langle x_f, t_f|\mathcal{T}\big[\hat{x}(t_1)\cdots\hat{x}(t_n)\big]|x_i, t_i\rangle, \tag{7.5}$$

where $\mathcal{T}$ is a sum over all $n!$ orderings, each multiplied by the corresponding product of step functions.

Remark 7.1. This is a *matrix element*, a transition amplitude between position eigenstates at different times. It is not an expectation value. Vacuum expectation values $\langle 0|\mathcal{T}[\cdots]|0\rangle$ are obtained by projecting onto the ground state, which is the job of complex time (Section 8.4).

7.4 Operator equations of motion [hw. pp. 22–24]

Take the harmonic oscillator, $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2x^2$. Classically $(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2)x = 0$. Quantum mechanically, the Heisenberg equations $\dot{\hat{x}} = \hat{p}/m$ and $\dot{\hat{p}} = -m\omega^2\hat{x}$ give the operator identity

$$\frac{\mathrm{d}^2\hat{x}}{\mathrm{d}t^2} = -\frac{i}{\hbar}\big[\dot{\hat{x}}(t), \hat{H}\big] = -\omega^2\hat{x}(t) \quad\Longrightarrow\quad \Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)\hat{x}(t) = 0. \tag{7.6}$$

Inside any matrix element the equation of motion holds, *provided no other insertion sits at the same time*. When one does, a contact term appears.

Operator computation. Using $\theta' = \delta$,

$$\partial_t \, \mathcal{T}[\hat{x}(t)\hat{x}(t')] = \delta(t-t') \, [\hat{x}(t), \hat{x}(t')] + \mathcal{T}[\hat{\dot{x}}(t)\hat{x}(t')], \tag{7.7}$$

$$\partial_t^2 \, \mathcal{T}[\hat{x}(t)\hat{x}(t')] = \partial_t \delta(t-t') \, [\hat{x}(t), \hat{x}(t')] + 2\,\delta(t-t') \, [\hat{\dot{x}}(t), \hat{x}(t')] + \mathcal{T}[\hat{\ddot{x}}(t)\hat{x}(t')]. \tag{7.8}$$

For the first term use (2.10) with the operator-valued $g(t) = [\hat{x}(t), \hat{x}(t')]$ at fixed t'. Since $g(t') = 0$ and $g'(t') = [\hat{\dot{x}}(t'), \hat{x}(t')] = [\hat{p}/m, \hat{x}] = -i\hbar/m$,

$$\partial_t \delta(t-t') \, [\hat{x}(t), \hat{x}(t')] = -g'(t') \, \delta(t-t') = +\frac{i\hbar}{m} \, \delta(t-t'). \tag{7.9}$$

The second term equals $2\,\delta(t-t')(-i\hbar/m)$. Adding,

$$\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)\mathcal{T}[\hat{x}(t)\hat{x}(t')] = \mathcal{T}\Big[\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)\hat{x}(t)\,\hat{x}(t')\Big] - \frac{i\hbar}{m}\delta(t-t') = -\frac{i\hbar}{m}\,\delta(t-t'). \tag{7.10}$$

$$\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)\langle f|\mathcal{T}[\hat{x}(t)\hat{x}(t')]|i\rangle = -\frac{i\hbar}{m}\,\delta(t-t')\,\langle f|i\rangle. \tag{7.11}$$

The right-hand side is zero classically and nonzero quantum mechanically. It is a *contact term*, proportional to $\hbar$.

Remark 7.2 (A shortcut). $\delta(t-t')[\hat{x}(t), \hat{x}(t')]$ vanishes identically as a distribution, by (2.9), since the equal-time commutator is zero. One may drop it at the first-derivative stage. The two routes agree, because in the longer route the $\partial_t\delta$ term, $+\frac{i\hbar}{m}\delta(t-t')$, cancels exactly one of the two $\delta(t-t')\,[\hat{\dot{x}}, \hat{x}]$ terms.

7.5 The same result from the path integral [hw. pp. 25–26]

Move the derivatives inside the path integral, which is allowed because t is an external label, and use (2.7):

$$\begin{aligned}\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2\Big)\int \mathcal{D}x\, e^{iS/\hbar}\, x(t)x(t') &= -\frac{1}{m}\int \mathcal{D}x\, e^{iS/\hbar}\,\frac{\delta S}{\delta x(t)}\,x(t') \\ &= \frac{i\hbar}{m}\int \mathcal{D}x\,\Big[\frac{\delta}{\delta x(t)}e^{iS/\hbar}\Big]x(t') \\ &= \frac{i\hbar}{m}\int \mathcal{D}x\,\Big\{\frac{\delta}{\delta x(t)}\Big[e^{iS/\hbar}x(t')\Big] - e^{iS/\hbar}\,\delta(t-t')\Big\}.\end{aligned} \tag{7.12}$$

The last step uses the Leibniz rule and (2.4).[2] The first term is the integral of a total functional derivative. If it vanishes,

$$\Big(\frac{\mathrm{d}^2}{\mathrm{d}t^2}+\omega^2\Big)\int \mathcal{D}x\, e^{iS/\hbar}\, x(t)x(t') = -\frac{i\hbar}{m}\,\delta(t-t')\int \mathcal{D}x\, e^{iS/\hbar}, \tag{7.13}$$

which agrees exactly with (7.11).

Does the total derivative really integrate to zero? [hw. pp. 25–26] In ordinary calculus $\int \mathrm{d}x\, f'(x) = 0$ when f vanishes at the boundary. Discretize to see what the "boundary" is here. With (2.3), $\delta/\delta x(t_n) \to \varepsilon^{-1}\partial/\partial x_n$, and the term becomes

$$\frac{1}{\varepsilon}\int \mathrm{d}x_n\, \frac{\partial}{\partial x_n}\Big[e^{iS_N/\hbar}x_{n'}\Big] = \frac{1}{\varepsilon}\Big[e^{iS_N/\hbar}x_{n'}\Big]_{x_n=-\infty}^{x_n=+\infty}. \tag{7.14}$$

The boundary is $|x_n| \to \infty$ for the *intermediate* points. The endpoints x_i, x_f are not integrated, so they are not boundaries. At $|x_n| \to \infty$ the integrand does not decay; it oscillates like $e^{imx_n^2/\hbar\varepsilon}$. As an ordinary limit the boundary term is undefined. This is the oscillatory-integral problem in its sharpest form. It is resolved by the damping supplied by complex time (Section 8.10).

7.6 The Schwinger–Dyson equation★

The computation above did not use any special property of the oscillator. For any action and any functional $F[x]$, once total functional derivatives integrate to zero,

$$0 = \int \mathcal{D}x\, \frac{\delta}{\delta x(t)}\Big[e^{iS/\hbar}F[x]\Big] = \int \mathcal{D}x\, e^{iS/\hbar}\Big[\frac{i}{\hbar}\frac{\delta S}{\delta x(t)}F + \frac{\delta F}{\delta x(t)}\Big], \tag{7.15}$$

that is,

Schwinger–Dyson equation

$$\Big\langle \frac{\delta S}{\delta x(t)}\,F[x]\Big\rangle = i\hbar\,\Big\langle \frac{\delta F}{\delta x(t)}\Big\rangle, \qquad \langle \mathcal{O}\rangle \equiv \int \mathcal{D}x\, e^{iS/\hbar}\,\mathcal{O}. \tag{7.16}$$

The classical equations of motion hold inside correlators up to contact terms of order $\hbar$. With $F = x(t')$ and $\delta S/\delta x(t) = -m(\partial_t^2+\omega^2)x(t)$ this is (7.11) in one line. The same equation, for fields, is derived in Peskin & Schroeder §9.6.

[2] The handwritten line on hw. p. 25 wrote the middle expression as $\frac{i\hbar}{m}\int \mathcal{D}x\, \frac{\delta}{\delta x(t)}\big[e^{iS/\hbar}x(t')\big]$ alone. That omits the $-e^{iS/\hbar}\delta(t-t')$ term, and with it the whole answer.

7.7 Time derivatives and coincident points★

The handwritten notes state [hw. p. 32] that "all other matrix elements can be obtained by taking time derivatives and coincident limits" of (7.5). That is true, with two caveats, and the computation of Section 7.4 is the prototype of both.

1. Derivatives do not commute with $\mathcal{T}$. A path-integral insertion of $\dot{x}$ computes the time derivative *of* the time-ordered product, not the time-ordered product of $\hat{\dot{x}}$. The two differ by contact terms. For example, differentiating $\mathcal{T}[\hat{x}(t)\hat{x}(t')]$ with respect to both times, as in Section 7.4,

$$\int \mathcal{D}x\, e^{iS/\hbar}\, \dot{x}(t)\, \dot{x}(t') = \partial_t \partial_{t'} \langle f|\mathcal{T}[\hat{x}(t)\hat{x}(t')]|i\rangle = \langle f|\mathcal{T}[\hat{\dot{x}}(t)\hat{\dot{x}}(t')]|i\rangle + \frac{i\hbar}{m}\,\delta(t-t')\,\langle f|i\rangle. \tag{7.17}$$

At coincident times the extra term is $\delta(0) \sim 1/\varepsilon$. This is the divergent "kinetic energy" of non-differentiable paths: by (4.14), $\langle(\Delta x)^2\rangle/\varepsilon^2 = i\hbar/m\varepsilon$.

2. Coincident products depend on discretization. The insertion $x(t)\dot{x}(t)$ has no unique meaning. Discretizing as $x_n(x_{n+1}-x_n)/\varepsilon$ or as $x_{n+1}(x_{n+1}-x_n)/\varepsilon$ gives results differing by $(x_{n+1}-x_n)^2/\varepsilon \to i\hbar/m$. These correspond to $\hat{p}\hat{x}/m$ and $\hat{x}\hat{p}/m$, whose difference is $[\hat{x},\hat{p}]/m = i\hbar/m$. A symmetric (midpoint) discretization gives the symmetrized product. This is the operator-ordering ambiguity of Section 5.2 seen from the configuration-space side.

8 Complex time: the Wick rotation and the $i\eta$ prescription

The handwritten notes list two difficulties in computing path integrals [hw. p. 33]: the integrals are oscillatory and ill-defined, and the measure needs fixing. They also name two cures for the first: Euclideanization (Wick rotation) and the $i\varepsilon$ prescription. This section explains *why* complexifying time works, from six complementary angles. It then sets up the Euclidean path integral systematically and shows that the two cures are one idea.

8.1 The problem: oscillation without decay

Every path contributes $e^{iS[x]/\hbar}$, of modulus one. No path is suppressed. The integral is held together only by cancellations, as in the Fresnel integral of Section 2.5. The same is visible in the energy basis,

$$e^{-i\hat{H}t/\hbar} = \sum_n e^{-iE_n t/\hbar} |n\rangle\langle n|. \tag{8.1}$$

Each coefficient is a unit phasor rotating at angular speed $E_n/\hbar$ (Fig. 13, left), and nothing ever decays. Sums and integrals of such terms converge at best conditionally. The manipulations we want (exchanging limits, dropping boundary terms, completing squares) are not justified.

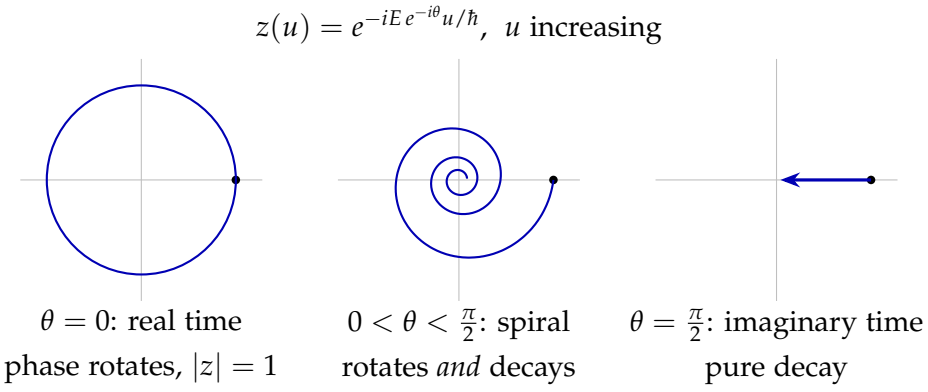


Figure 13: The amplitude $e^{-iEt/\hbar}$ traced in the complex plane as $|t|$ grows along the ray $t = e^{-i\theta}u$. Real time gives an endless circle. Any clockwise tilt of the time axis turns it into an inward spiral. At $\theta = \pi/2$ the oscillation is replaced entirely by exponential decay.

8.2 Intuition 1: rotate into the direction of decay

> **Intuition: the Fresnel integral, again**
>
> In Section 2.5 the non-decaying integral $\int e^{iax^2}\mathrm{d}x$ was computed by noticing that e^{iaz^2} is analytic and *decays* along the ray $z = se^{i\pi/4}$. Cauchy's theorem let us rotate the contour there without changing the answer (Fig. 3).

Apply the same idea to time. As a function of complex t,

$$\left|e^{-iEt/\hbar}\right| = e^{E\,\mathrm{Im}\,t/\hbar}, \tag{8.2}$$

which decays for $\mathrm{Im}\,t < 0$ when $E > 0$. The analogue of rotating x by $\pi/4$ is rotating t *clockwise* into the lower half-plane. The recipe is the same as for Fresnel:

1. find the direction in which the integrand decays;
2. use analyticity to rotate the contour there;
3. compute the now well-defined integral;
4. rotate back.

8.3 Intuition 2: the propagator lives in the lower half t-plane

Complexify time as $t = e^{-i\theta}u$ with $u > 0$ [hw. p. 34]. Since $-ie^{-i\theta} = -i\cos\theta - \sin\theta$, each spectral term becomes

$$e^{-iE_nt/\hbar} = e^{-iE_n\cos\theta\,u/\hbar}\;e^{-E_n\sin\theta\,u/\hbar}. \tag{8.3}$$

For $0 < \theta < \pi$ this decays, provided the spectrum is **bounded below**. Strictly positive energies are not needed: factor out $e^{-iE_0t/\hbar}$ and all *relative* weights $e^{-(E_n-E_0)\sin\theta\,u/\hbar}$ decay.[3] Therefore

$$K(t) = \sum_n \psi_n(x_f)\psi_n^*(x_i)\,e^{-iE_nt/\hbar} \tag{8.4}$$

converges better off the real axis, and defines an *analytic function of t in the open lower half-plane*. The physical, real-time propagator is its boundary value,

$$K(t) = \lim_{\theta\to 0^+} K\big(e^{-i\theta}t\big). \tag{8.5}$$

Why clockwise? Because the spectrum is bounded *below* and unbounded *above*. Rotating counterclockwise ($\mathrm{Im}\,t > 0$) would make the high-energy terms blow up.

[3] This corrects "$E_n > 0$ for all n, a bounded system" on hw. p. 34. Shifting $\hat{H}$ by a constant changes nothing, and the free particle, with its continuous spectrum $E \geq 0$, is fine.

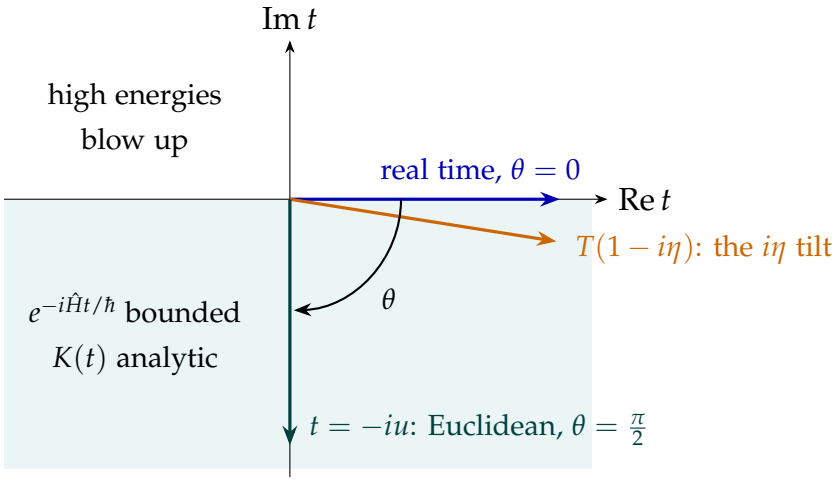


Figure 14: The complex time plane. The propagator is analytic in the shaded lower half-plane, because $\hat{H}$ is bounded below. Real time is its upper edge. The $i\eta$ prescription is an infinitesimal clockwise tilt; the Wick rotation is the full quarter turn to the negative imaginary axis. Both are the same analytic continuation.

Remark 8.1 (Energy plane versus time plane). In the energy representation (6.7) the same continuation appears as a *counterclockwise* rotation of the energy contour, $E = iE_{\mathrm{E}}$. With energies measured so that $E_n > 0$, this avoids the poles at $E_n - i\eta$. Time rotates clockwise and energy counterclockwise, so the product $Et = (iE_{\mathrm{E}})(-iu) = E_{\mathrm{E}}u$ stays real. This is the same rotation $p^0 \to ip^0_{\mathrm{E}}$ used for Feynman propagators in quantum field theory.

8.4 Intuition 3: imaginary time projects onto the ground state

At $\theta = \pi/2$, with $\tau \equiv u$,

$$e^{-\hat{H}\tau/\hbar}|\psi\rangle = \sum_n c_n\, e^{-E_n\tau/\hbar}|n\rangle = e^{-E_0\tau/\hbar}\Big[c_0|0\rangle + \sum_{n\geq 1} c_n\, e^{-(E_n-E_0)\tau/\hbar}|n\rangle\Big]$$
$$\xrightarrow{\tau\to\infty}\; c_0\, e^{-E_0\tau/\hbar}|0\rangle. \tag{8.6}$$

Imaginary-time evolution is cooling. Every excited component dies faster than the ground state (Fig. 15). Two consequences follow.

1. For large τ, $\langle x_f|e^{-\hat{H}\tau/\hbar}|x_i\rangle \simeq \psi_0(x_f)\psi_0^*(x_i)\, e^{-E_0\tau/\hbar}$. The ground-state energy and wavefunction can be read off a Euclidean path integral. This is how lattice and quantum Monte Carlo calculations work.

2. Vacuum correlators. The matrix elements of Section 7 become vacuum expectation values:

$$\langle 0|\mathcal{T}[\hat{x}(t_1)\cdots\hat{x}(t_n)]|0\rangle = \lim_{T\to\infty(1-i\eta)} \frac{\langle x_f, T|\mathcal{T}[\hat{x}(t_1)\cdots\hat{x}(t_n)]|x_i, -T\rangle}{\langle x_f, T|x_i, -T\rangle}. \tag{8.7}$$

This is the quantum-mechanical version of the Gell-Mann–Low formula (Peskin & Schroeder: §4.2 in operator form, §9.2 in path-integral form). The infinitesimal tilt $T \to T(1 - i\eta)$ supplies factors $e^{-\eta E_n T/\hbar}$ that select $|0\rangle$ at both ends.

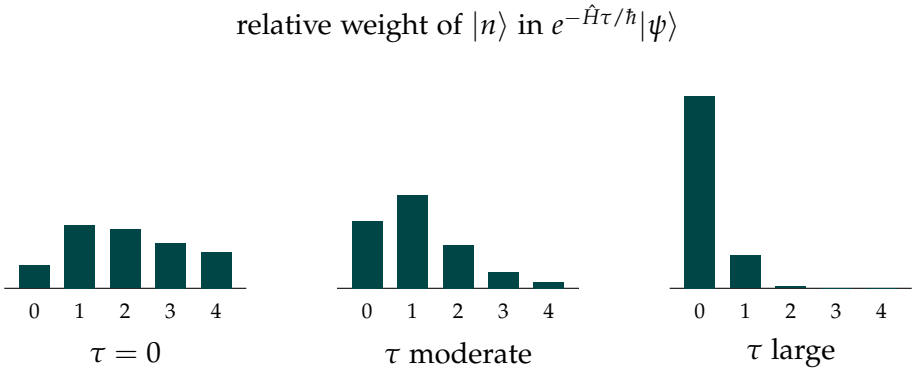


Figure 15: Imaginary-time evolution suppresses each level by $e^{-(E_n - E_0)\tau/\hbar}$ relative to the ground state. A generic state (left) is "cooled" into $|0\rangle$ (right).

8.5 Intuition 4: why time ordering is forced

Consider a time-ordered matrix element with $t_f > t_1 > \cdots > t_n > t_i$ [hw. p. 34]:

$$\langle x_f | e^{-i\hat{H}(t_f - t_1)/\hbar}\, \hat{O}_1\, e^{-i\hat{H}(t_1 - t_2)/\hbar}\, \hat{O}_2 \cdots \hat{O}_n\, e^{-i\hat{H}(t_n - t_i)/\hbar} | x_i \rangle. \tag{8.8}$$

Every elapsed time Δt_k is *positive*, so after $t \to e^{-i\theta}u$ every evolution factor is damped by (8.3). In a product that is not time-ordered, some factor is $e^{+i\hat{H}|\Delta t|/\hbar}$, which becomes $e^{+\hat{H}\sin\theta|\Delta u|/\hbar}$ and is unbounded, because $\hat{H}$ is unbounded above. *Only time-ordered products can be continued to imaginary time.* Conversely, Euclidean correlators continue back to time-ordered real-time ones. The time ordering found in Section 7 is not an accident of the path integral; it is the condition for analyticity.

8.6 Intuition 5: geometry, hyperbolae become circles [hw. p. 35]

In $1+1$ dimensions with $t = -iu$, the interval becomes

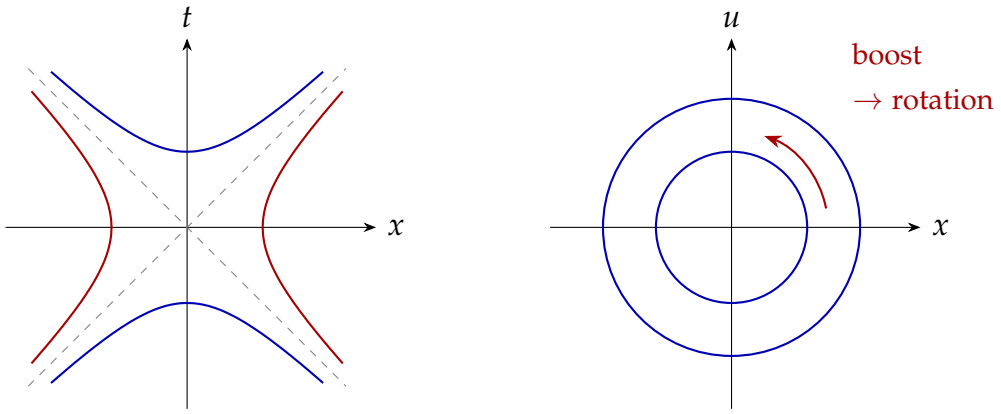
$$\mathrm{d}s^2 = \mathrm{d}t^2 - \mathrm{d}x^2 \;\longrightarrow\; -(\mathrm{d}u^2 + \mathrm{d}x^2). \tag{8.9}$$

In the $(-,+)$ signature used on hw. p. 35 this reads $-\mathrm{d}t^2 + \mathrm{d}x^2 \to \mathrm{d}u^2 + \mathrm{d}x^2$. Up to an overall sign the metric is Euclidean, hence the name. More precisely:

- Surfaces of constant interval, the hyperbolae $t^2 - x^2 = \text{const}$, become circles $u^2 + x^2 = \text{const}$.

- Boosts become rotations. With rapidity $\chi = i\phi$, $\cosh\chi = \cos\phi$ and $\sinh\chi = i\sin\phi$, the boost $t' = t\cosh\chi - x\sinh\chi$, $x' = x\cosh\chi - t\sinh\chi$ becomes $u' = u\cos\phi + x\sin\phi$, $x' = x\cos\phi - u\sin\phi$.
- The light cone $t = \pm x$ has no real Euclidean image: $u^2 + x^2 = 0$ only at the origin. There is no causal structure; all directions are equivalent.

Real-time oscillation becomes Euclidean decay, and propagation becomes correlation that falls off with distance. Causality is invisible in Euclidean signature and reappears on rotating back.



Minkowski: $t^2 - x^2 = \pm 1$, light cone dashed

Euclidean: $u^2 + x^2 = \text{const}$

Figure 16: Under $t = -iu$, invariant hyperbolae become circles, boosts become rotations, and the light cone collapses to a point.

8.7 Intuition 6: quantum mechanics in imaginary time is statistical mechanics

The polymer picture. The discretized Euclidean action (derived in Section 8.8) is

$$\frac{S_{\mathrm{E}}}{\hbar} = \frac{1}{\hbar}\sum_n \left[\frac{m}{2\varepsilon}(x_{n+1} - x_n)^2 + \varepsilon\, V(x_n)\right]. \tag{8.10}$$

Compare the Boltzmann weight $e^{-E_{\mathrm{config}}/k_B T}$ of a classical system: the weight $e^{-S_{\mathrm{E}}/\hbar}$ describes a *necklace* of beads at positions x_n, joined by springs of stiffness m/ε, each bead sitting in the potential εV, at "temperature" $k_B T \leftrightarrow \hbar$ (Fig. 17).

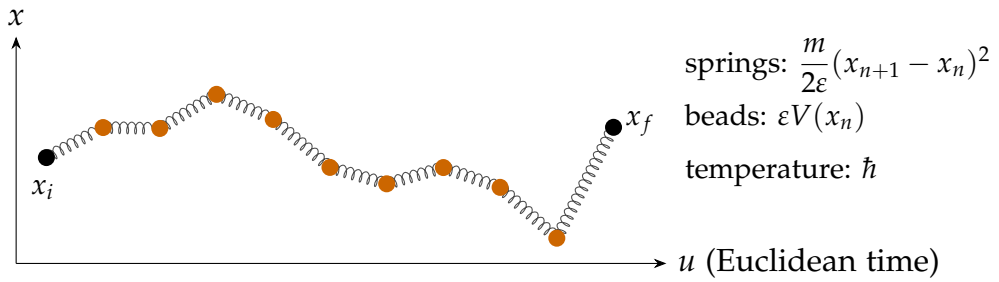


Figure 17: The Euclidean path integral as a classical polymer. Quantum fluctuations of the particle are thermal fluctuations of the necklace. As $\hbar \to 0$ the necklace freezes into its minimum-energy shape, the Euclidean classical path.

quantum mechanics in imaginary time	classical statistical mechanics
Euclidean path $\bar{x}(u)$	configuration of a polymer
$S_{\mathrm{E}}[\bar{x}]$	energy of the configuration
$\hbar$	temperature $k_B T$
$\hbar \to 0$ (classical limit)	zero temperature (ground configuration)
Tr $e^{-\beta \hat{H}}$	ring polymer of length $\beta\hbar$ (Section 8.11)

The diffusion picture. Substituting $t = -iu$ ($\partial_t = i\partial_u$) into the Schrödinger equation gives

$$\hbar\, \partial_u \psi = \frac{\hbar^2}{2m} \partial_x^2 \psi - V(x)\psi. \tag{8.11}$$

This is a *diffusion equation* with diffusion constant $D = \hbar/2m$ and absorption rate $V/\hbar$. The free Euclidean propagator is the heat kernel,

$$K_{\mathrm{E}}^{\mathrm{free}}(x_f, x_i; \tau) = \sqrt{\frac{m}{2\pi\hbar\tau}}\; e^{-m(x_f - x_i)^2/2\hbar\tau}, \tag{8.12}$$

which is real, positive and normalized. It is the transition probability of a Brownian particle. Its typical displacement $\sqrt{\hbar\tau/m}$ matches (4.14): the jagged, nowhere differentiable paths of Fig. 7 are *Brownian paths*. In imaginary time, $e^{-S_{\mathrm{E}}/\hbar}\mathcal{D}x$ is an honest probability measure (Wiener measure), which is why the Euclidean path integral is mathematically rigorous while the real-time one is not. With a potential, the Feynman–Kac formula expresses $\langle x_f | e^{-\hat{H}\tau/\hbar} | x_i \rangle$ as $K_{\mathrm{E}}^{\mathrm{free}}$ times the average of $\exp\big[-\frac{1}{\hbar}\int_0^\tau V(\bar{x}(u))\,\mathrm{d}u\big]$ over Brownian bridges from x_i to x_f.

The inverted potential. The Euclidean equation of motion is $\delta S_{\mathrm{E}}/\delta\bar{x} = -m\bar{x}'' + V'(\bar{x}) = 0$, that is, $m\bar{x}'' = +V'(\bar{x})$: Newton's law in the potential $-V$. Classically forbidden regions become allowed, which is how tunnelling and instantons enter.

8.8 Systematics of the Euclidean path integral [hw. pp. 35–36]

Define Euclidean paths $\bar{x}(u) \equiv x(t = -iu)$. Then

$$\mathrm{d}t = -i\,\mathrm{d}u, \qquad \dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t} = i\frac{\mathrm{d}\bar{x}}{\mathrm{d}u} \equiv i\bar{x}'. \tag{8.13}$$

The real-time weight becomes

$$\begin{gathered}\frac{i}{\hbar}\int \mathrm{d}t\, L(x,\dot{x}) = \frac{i}{\hbar}\int (-i\,\mathrm{d}u)\, L(\bar{x}, i\bar{x}') = -\frac{1}{\hbar}\int \mathrm{d}u\, L_{\mathrm{E}}(\bar{x},\bar{x}'), \\ L_{\mathrm{E}}(\bar{x},\bar{x}') \equiv -L(\bar{x}, i\bar{x}').\end{gathered} \tag{8.14}$$

Euclidean path integral

$$K_{\mathrm{E}}(x_f, x_i; \tau) \equiv \langle x_f | e^{-\hat{H}\tau/\hbar} | x_i \rangle = \int_{\bar{x}(0)=x_i}^{\bar{x}(\tau)=x_f} \mathcal{D}\bar{x}\; e^{-S_{\mathrm{E}}[\bar{x}]/\hbar}, \qquad S_{\mathrm{E}} = \int_0^\tau \mathrm{d}u\, L_{\mathrm{E}}. \tag{8.15}$$

For $L = \frac{1}{2}m\dot{x}^2 - V(x)$:

$$L_{\mathrm{E}} = -\left[\tfrac{1}{2}m(i\bar{x}')^2 - V(\bar{x})\right] = \tfrac{1}{2}m\bar{x}'^2 + V(\bar{x}). \tag{8.16}$$

The real-time propagator is recovered by continuing back, $\tau \to iT$, i.e. $K(T) = K_{\mathrm{E}}(\tau = iT)$.

For V bounded below, S_{E} is bounded below. The integral is dominated by the *minima* of S_{E}, which are genuine minima in Euclidean signature, and large deviations are exponentially suppressed. (On hw. p. 36 the substitution was written $t = iu$; it should read $t = -iu$, as on hw. p. 35.)

8.9 The $i\eta$ prescription, and why it is the same thing [hw. p. 36]

The handwritten notes add a term to the real-time Lagrangian:

$$L \to L + \frac{i\eta}{2}x^2 \quad \Longrightarrow \quad e^{iS/\hbar} \to e^{iS/\hbar}\, e^{-\frac{\eta}{2\hbar}\int \mathrm{d}t\, x^2}, \qquad \eta \to 0^+. \tag{8.17}$$

This is the quantum-mechanical analogue of the field-theory shift $m^2 \to m^2 - i\eta$. Each intermediate-point integral acquires a Gaussian damping factor and becomes absolutely convergent. Why is this the same as tilting time? For the oscillator, the added term shifts $m\omega^2 \to m\omega^2 - i\eta$, so $\omega \to \omega - i\eta'$ with $\eta' = \eta/2m\omega$. The phases of the energy eigenstates become

$$e^{-i\omega(n+\frac{1}{2})T} \to e^{-i\omega(n+\frac{1}{2})T}\, e^{-\eta'(n+\frac{1}{2})T}, \tag{8.18}$$

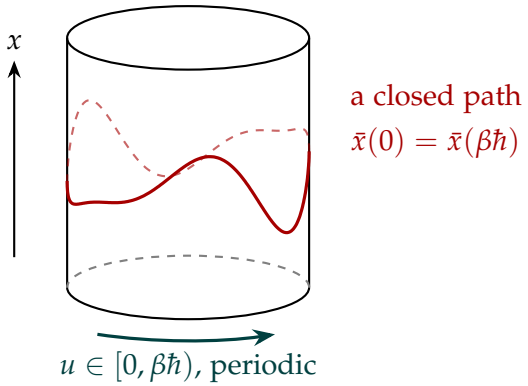


Figure 18: The thermal trace is a path integral over closed paths on a cylinder whose circumference is the Euclidean time $\beta\hbar$. In the polymer picture it is a ring polymer.

which is *exactly* the damping produced by $T \to T(1 - i\eta'/\omega)$. Both prescriptions choose the same branch of every square root and the same analytic continuation. One is the infinitesimal version of the other (Fig. 14).

> **Caution: continuing back**
>
> Continuation from τ back to iT is unique when the function is known analytically, which is guaranteed by $\hat{H}$ being bounded below. Numerically, reconstructing real-time data from noisy Euclidean data is an ill-conditioned inverse problem. That is why lattice methods excel at static quantities and struggle with real-time dynamics.

8.10 Resolution of the boundary question of Section 7.5

With either regulator, the dependence of the integrand on an intermediate point x_n is, for large $|x_n|$,

$$\text{Euclidean: } e^{-\frac{m}{2\hbar\varepsilon}[(x_{n+1}-x_n)^2+(x_n-x_{n-1})^2]} \sim e^{-mx_n^2/\hbar\varepsilon}, \qquad i\eta\text{: } e^{imx_n^2/\hbar\varepsilon}\, e^{-\eta\varepsilon x_n^2/2\hbar}. \tag{8.19}$$

Both decay like Gaussians, so the boundary term $[e^{iS_N/\hbar}x_{n'}]_{x_n=-\infty}^{+\infty}$ vanishes, for every N and every $\eta > 0$. The Schwinger–Dyson equation (7.16) therefore holds exactly in the regulated theory, and survives the limits $N \to \infty$ and $\eta \to 0^+$. *Total functional derivatives integrate to zero precisely because the path integral is defined by analytic continuation.*

8.11 The partition function: periodic imaginary time [hw. p. 37]

Setting $\tau = \beta\hbar$ and taking the trace,

$$Z(\beta) = \operatorname{Tr} e^{-\beta\hat{H}} = \sum_n e^{-\beta E_n} = \int \mathrm{d}x\, K_{\mathrm{E}}(x, x; \beta\hbar) = \oint_{\bar{x}(0)=\bar{x}(\beta\hbar)} \mathcal{D}\bar{x}\, e^{-S_{\mathrm{E}}[\bar{x}]/\hbar}. \tag{8.20}$$

The trace sets the final point equal to the initial one and integrates over it. Z is a sum over *closed* paths in Euclidean time of period $\beta\hbar$. Euclidean time has become a circle (Fig. 18). The real-time trace Tr $e^{-i\hat{H}t/\hbar}$ is not yet a partition function; it becomes one at $t = -i\beta\hbar$.[4]

Remark 8.2 (Where this leads). "Finite temperature = periodic imaginary time" is the foundation of thermal field theory. In gravity it underlies the Euclidean (Gibbons–Hawking) approach to black-hole thermodynamics. Requiring the Euclidean black-hole geometry to be smooth at the horizon fixes the period of imaginary time, and hence the Hawking temperature.

[4]Hw. p. 37 wrote $t = i\hbar\beta$. Check: $e^{-i\hat{H}(-i\beta\hbar)/\hbar} = e^{-\beta\hat{H}}$.

9 Computing path integrals: the harmonic oscillator

9.1 The measure problem [hw. p. 37]

The left-hand side of $\int \mathcal{D}x\, e^{iS[x]/\hbar} = \langle x_f | e^{-i\hat{H}T/\hbar} | x_i \rangle$ depends on the normalization $\mathcal{N}_N$ hidden in $\mathcal{D}x$ (2.1); the right-hand side is unambiguous. One exact calculation fixes the normalization. Section 5.3 showed that $\mathcal{N}_N = A^N$ with $A = \sqrt{m/2\pi i\hbar\varepsilon}$, which depends on the *kinetic term only*, not on V. The free particle therefore fixes the measure for every potential with the same kinetic term, in particular the oscillator. (A different kinetic term, such as a position-dependent mass or a curved configuration space, changes the measure; see Section 5.4.)

9.2 Expansion about the classical path [hw. p. 38]

Let $S[x] = \frac{m}{2}\int_{t_i}^{t_f} (\dot{x}^2 - \omega^2 x^2)\,\mathrm{d}t$ and write

$$x(t) = x_{\mathrm{cl}}(t) + y(t), \qquad y(t_i) = y(t_f) = 0, \tag{9.1}$$

where x_{cl} is the classical solution with the given endpoints. The shift is a translation, so $\mathcal{D}x = \mathcal{D}y$. Expanding,

$$S[x_{\mathrm{cl}} + y] = S[x_{\mathrm{cl}}] + m\int \big(\dot{x}_{\mathrm{cl}}\dot{y} - \omega^2 x_{\mathrm{cl}} y\big)\,\mathrm{d}t + \frac{m}{2}\int \big(\dot{y}^2 - \omega^2 y^2\big)\,\mathrm{d}t. \tag{9.2}$$

The linear term equals $[m\dot{x}_{\mathrm{cl}}y]_{t_i}^{t_f} - m\int(\ddot{x}_{\mathrm{cl}} + \omega^2 x_{\mathrm{cl}})y\,\mathrm{d}t = 0$, by the boundary conditions and the equation of motion. Integrating the quadratic term by parts,

$$S[x_{\mathrm{cl}} + y] = S_{\mathrm{cl}} + \frac{m}{2}\int_{t_i}^{t_f} \mathrm{d}t\; y\Big(-\partial_t^2 - \omega^2\Big)y \qquad \text{(exact)}. \tag{9.3}$$

For a quadratic action there are no cubic or higher terms, so the stationary-phase evaluation is exact (Section 2.5):

$$K(f;i) = e^{iS_{\mathrm{cl}}/\hbar}\,F(T), \qquad F(T) = \int_{y(t_i)=y(t_f)=0} \mathcal{D}y\;\exp\Big[\frac{i}{\hbar}\,\frac{m}{2}\int \mathrm{d}t\; y\big(-\partial_t^2 - \omega^2\big)y\Big]. \tag{9.4}$$

F depends only on the elapsed time T, not on x_i or x_f, because the fluctuation integral knows nothing about the endpoints.

9.3 The fluctuation integral by mode expansion [hw. pp. 39–40]

Because F depends only on T, we may compute it in imaginary time and continue back ($T = -i\tau$; the classical phase $e^{iS_{\mathrm{cl}}/\hbar}$ is kept in real time, which is legitimate for exactly this reason):

$$F_{\mathrm{E}}(\tau) = \int \mathcal{D}\bar{y}\;\exp\Big[-\frac{m}{2\hbar}\int_0^{\tau} \mathrm{d}u\;\bar{y}\big(-\partial_u^2 + \omega^2\big)\bar{y}\Big]. \tag{9.5}$$

Expand in the eigenfunctions of $-\partial_u^2$ that vanish at both ends (Fig. 19):

$$\bar{y}(u) = \sum_{n\geq 1} \tilde{y}_n \, s_n(u), \qquad s_n(u) = \sqrt{\frac{2}{\tau}} \sin \frac{n\pi u}{\tau}, \qquad -\partial_u^2 s_n = \lambda_n s_n, \quad \lambda_n = \frac{n^2\pi^2}{\tau^2}. \tag{9.6}$$

The s_n are orthonormal on $[0,\tau]$, so the action is diagonal: $\int_0^\tau \bar{y}(-\partial_u^2 + \omega^2)\bar{y}\,\mathrm{d}u = \sum_n (\lambda_n + \omega^2)\tilde{y}_n^2$.

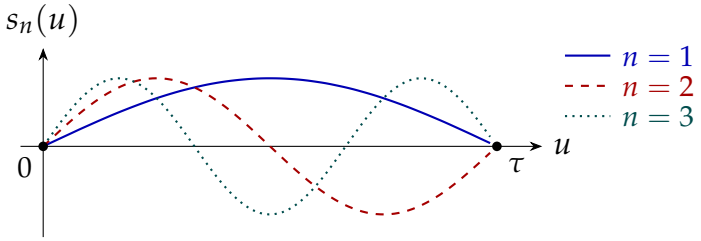


Figure 19: Fluctuation modes with fixed endpoints, $y(0) = y(\tau) = 0$. Each mode is an independent Gaussian integration variable.

The change of variables from the time-slice values $\{y_k\}$ to the mode amplitudes $\{\tilde{y}_n\}$ is orthogonal (a discrete sine transform), so its Jacobian C is a constant *independent of* ω. It may depend on N, τ, m and $\hbar$. This independence is the whole trick: everything ω-independent can be fixed later by the free particle. The integral is a product of decoupled Gaussians:

$$F_{\mathrm{E}}(\tau) = C \prod_{n\geq 1} \sqrt{\frac{2\pi\hbar}{m(\lambda_n + \omega^2)}} = \underbrace{C \prod_{n\geq 1} \sqrt{\frac{2\pi\hbar}{m\lambda_n}}}_{\equiv N_{\mathrm{E}}(\tau)} \prod_{n\geq 1} \left(1 + \frac{\omega^2}{\lambda_n}\right)^{-1/2}. \tag{9.7}$$

In Euclidean signature the factor is $1 + \omega^2/\lambda_n$, with a plus sign. The minus sign of the handwritten first line on p. 40 appears only after continuing back. The prefactor N_{E} diverges, but it is F_{E} at $\omega = 0$, the free particle. The remaining product converges, since $\sum_n \omega^2/\lambda_n < \infty$. Using Euler's product $\frac{\sinh z}{z} = \prod_{n\geq 1}\left(1 + \frac{z^2}{n^2\pi^2}\right)$ with $z = \omega\tau$,

$$F_{\mathrm{E}}(\tau) = N_{\mathrm{E}}(\tau) \sqrt{\frac{\omega\tau}{\sinh \omega\tau}}. \tag{9.8}$$

Continuing $\tau \to iT$ turns $\lambda_n \to -n^2\pi^2/T^2$ and $\sinh(i\omega T) = i \sin \omega T$, so

$$F(T) = N(T) \prod_{n\geq 1} \left(1 - \frac{\omega^2 T^2}{n^2\pi^2}\right)^{-1/2} = N(T) \sqrt{\frac{\omega T}{\sin \omega T}}, \tag{9.9}$$

by the companion product $\frac{\sin z}{z} = \prod_{n\geq 1}\left(1 - \frac{z^2}{n^2\pi^2}\right)$. Here $N(T)$ is the free-particle prefactor.

Remark 9.1 (Determinants and the Gelfand–Yaglom method). What was computed is a ratio of determinants, $F(\omega)/F(0) = \big[\det(-\partial^2+\omega^2)/\det(-\partial^2)\big]^{-1/2}$. It is finite even though each determinant diverges. When the eigenvalues are not known in closed form (anharmonic potentials, fluctuations around a non-trivial saddle), the Gelfand–Yaglom theorem gives the ratio by solving an ODE: $\det(-\partial^2+W)/\det(-\partial^2) = \psi_W(\tau)/\psi_0(\tau)$, where $(-\partial^2+W)\psi_W = 0$ with $\psi_W(0)=0$ and $\psi_W'(0)=1$. For $W=\omega^2$, $\psi_W = \sinh(\omega u)/\omega$ and $\psi_0 = u$, which reproduces $\sinh\omega\tau/\omega\tau$.

9.4 Fixing the normalization with the free particle [hw. p. 41]

Compute the free propagator directly in Schrödinger theory. It is convenient to work in imaginary time, where the momentum integral is an ordinary Gaussian:

$$K_{\mathrm{E}}^{\mathrm{free}} = \langle x_f | e^{-\tau\hat{p}^2/2m\hbar} | x_i\rangle = \int \frac{\mathrm{d}p}{2\pi\hbar}\, e^{ip(x_f-x_i)/\hbar - \tau p^2/2m\hbar} = \sqrt{\frac{m}{2\pi\hbar\tau}}\; e^{-m(x_f-x_i)^2/2\hbar\tau}. \tag{9.10}$$

Continuing $\tau \to iT$ (or using the Fresnel formula directly in real time):

$$\boxed{K_{\mathrm{free}}(f;i) = \sqrt{\frac{m}{2\pi i\hbar T}}\; \exp\Big[\frac{i}{\hbar}\frac{m(x_f-x_i)^2}{2T}\Big]} \tag{9.11}$$

The exponent is S_{cl} for uniform motion. Comparing with (9.4) at $\omega = 0$ fixes the prefactor:

$$N(T) = \sqrt{\frac{m}{2\pi i\hbar T}}, \qquad N_{\mathrm{E}}(\tau) = \sqrt{\frac{m}{2\pi\hbar\tau}}. \tag{9.12}$$

The divergent product $C\prod_n \sqrt{2\pi\hbar/m\lambda_n}$ [hw. pp. 40–41] has been replaced by this finite answer. The infinite constant C absorbs the divergence.

9.5 The harmonic-oscillator propagator [hw. pp. 41–42]

Combining (9.4), (9.9) and $N(T)$, and using the classical action derived in Appendix A:

Harmonic-oscillator propagator ($0 < \omega T < \pi$)

$$K_{\mathrm{HO}}(f;i) = \sqrt{\frac{m\omega}{2\pi i\hbar \sin\omega T}}\; \exp\Big\{\frac{im\omega}{2\hbar\sin\omega T}\Big[(x_i^2+x_f^2)\cos\omega T - 2x_i x_f\Big]\Big\}. \tag{9.13}$$

In imaginary time ($T=-i\tau$):

$$K_{\mathrm{E}}^{\mathrm{HO}} = \sqrt{\frac{m\omega}{2\pi\hbar \sinh\omega\tau}}\; \exp\Big\{-\frac{m\omega}{2\hbar\sinh\omega\tau}\Big[(x_i^2+x_f^2)\cosh\omega\tau - 2x_i x_f\Big]\Big\}. \tag{9.14}$$

9.6 Beyond the first focal point: the Maslov phase★

For $\omega T > \pi$, the factors $1 - \omega^2 T^2/n^2\pi^2$ in (9.9) with $n\pi < \omega T$ are negative, and the square root is ambiguous. The $i\eta$ prescription resolves the ambiguity. With $\omega^2 \to \omega^2 - i\eta$, each negative factor becomes (negative) $+\,i0$, just above the negative real axis, and contributes $(\cdot)^{-1/2} = |\cdot|^{-1/2} e^{-i\pi/2}$. Equivalently, in the real-time mode expansion the quadratic form $\int y(-\partial_t^2 - \omega^2)y$ has eigenvalues $(n\pi/T)^2 - \omega^2$. Each negative eigenvalue turns a Fresnel factor $e^{+i\pi/4}$ into $e^{-i\pi/4}$, a relative phase $e^{-i\pi/2}$ (Section 2.5). These negative modes are exactly the directions in which the classical path stops being a minimum (3.3). The result, valid for $\omega T \notin \pi\mathbb{Z}$, is

$$K_{\mathrm{HO}}(f;i) = \sqrt{\frac{m\omega}{2\pi i\hbar\,|\sin\omega T|}}\; e^{-i\pi\mu/2}\, e^{iS_{\mathrm{cl}}/\hbar}, \qquad \mu = \left\lfloor \frac{\omega T}{\pi} \right\rfloor, \tag{9.15}$$

where μ, the *Maslov index*, counts the conjugate points passed.

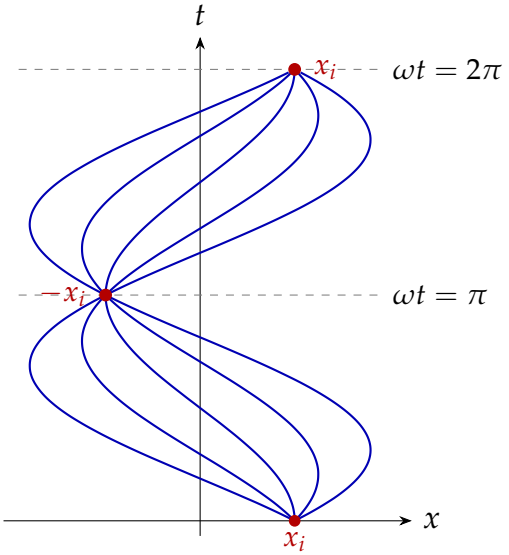


Figure 20: Classical oscillator trajectories leaving x_i with different momenta, $x_{\mathrm{cl}}(t) = x_i \cos\omega t + \frac{p_i}{m\omega}\sin\omega t$, all refocus at $-x_i$ when $\omega t = \pi$ and at x_i when $\omega t = 2\pi$. These are the conjugate points. Each one passed adds $e^{-i\pi/2}$ to the propagator.

At the focal points themselves. As $\omega T \to \pi$, the prefactor diverges and the Gaussian in $x_f + x_i$ narrows. One finds $K \to e^{-i\pi/2}\,\delta(x_f + x_i)$: after half a period every path has been mapped to $-x_i$. This agrees with the spectral decomposition, $e^{-i\hat{H}\pi/\hbar\omega} = e^{-i\pi(\hat{n}+\frac{1}{2})} = e^{-i\pi/2}(-1)^{\hat{n}} = e^{-i\pi/2}\hat{P}$, where $\hat{P}$ is parity. At $\omega T = 2\pi$, $K = -\delta(x_f - x_i)$. The overall -1 after a full period is the zero-point phase $e^{-i\omega T/2}$.

9.7 Checks★

1. **Free limit.** As $\omega \to 0$, $\sin\omega T \to \omega T$ and $\cos\omega T \to 1$, and (9.13) reduces to (9.11).

2. **Short times.** As $T \to 0$, $K_{\rm HO} \to \sqrt{m/2\pi i\hbar T}\, e^{im(x_f-x_i)^2/2\hbar T} \to \delta(x_f - x_i)$.

3. **Schrödinger equation.** (9.13) satisfies (6.3) with $H = -\frac{\hbar^2}{2m}\partial_x^2 + \frac{1}{2}m\omega^2 x^2$ (a good exercise).

4. **The spectrum, from the thermal trace.** Set $x_i = x_f = x$ and $\tau = \beta\hbar$ in (9.14). Using $(\cosh\omega\tau - 1)/\sinh\omega\tau = \tanh(\omega\tau/2)$,

$$K_{\rm E}^{\rm HO}(x,x;\tau) = \sqrt{\frac{m\omega}{2\pi\hbar\sinh\omega\tau}}\; e^{-\frac{m\omega}{\hbar}\tanh(\omega\tau/2)\,x^2}. \tag{9.16}$$

   Integrating over x (8.20) and using $\sinh\omega\tau\,\tanh(\omega\tau/2) = 2\sinh^2(\omega\tau/2)$:

$$\begin{aligned} Z(\beta) &= \frac{1}{\sqrt{2\sinh(\omega\tau)\tanh(\omega\tau/2)}} = \frac{1}{2\sinh(\beta\hbar\omega/2)} \\ &= \frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}} = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+\frac{1}{2})}. \end{aligned} \tag{9.17}$$

   Comparing with $Z = \sum_n e^{-\beta E_n}$ gives $E_n = \hbar\omega\big(n+\frac{1}{2}\big)$, including the zero-point energy. This is the strongest possible check: the measure, the determinant and the continuation are all correct.

A The classical action of the harmonic oscillator [hw. p. 42]

Measure time from $t_i = 0$. The solution of $\ddot{x} + \omega^2 x = 0$ with $x(0) = x_i$ and $x(T) = x_f$ is

$$x_{\rm cl}(t) = \frac{x_i \sin\omega(T-t) + x_f \sin\omega t}{\sin\omega T}, \tag{A.1}$$

which satisfies both boundary conditions by inspection. On a solution, $\frac{\rm d}{{\rm d}t}(x\dot{x}) = \dot{x}^2 + x\ddot{x} = \dot{x}^2 - \omega^2 x^2$, which is $2L/m$. The action is therefore a pure boundary term:

$$S_{\rm cl} = \frac{m}{2}\int_0^T (\dot{x}^2 - \omega^2 x^2)\,{\rm d}t = \frac{m}{2}\Big[x\dot{x}\Big]_0^T = \frac{m}{2}\big(x_f\dot{x}(T) - x_i\dot{x}(0)\big). \tag{A.2}$$

With $\dot{x}(T) = \omega\,\frac{x_f\cos\omega T - x_i}{\sin\omega T}$ and $\dot{x}(0) = \omega\,\frac{x_f - x_i\cos\omega T}{\sin\omega T}$,

$$\boxed{S_{\rm cl} = \frac{m\omega}{2\sin\omega T}\Big[(x_i^2 + x_f^2)\cos\omega T - 2x_i x_f\Big]} \tag{A.3}$$

As checks: $\omega \to 0$ gives $m(x_f - x_i)^2/2T$, and $\partial S_{\rm cl}/\partial x_f = m\dot{x}(T) = p_f$, in agreement with (3.5).

B Map of the handwritten notes and correction log

B.1 Where each handwritten page went

handwritten pp.	content	here
1	motivation: Schrödinger vs. Lagrangian formulations	§1
2	functions vs. functionals; properties of the action	§2.1, §3
3	oscillator phase space; Dirac's proposal	§3.4, §4.2
4–5	additivity, time slicing, Feynman's suggestion	§4.4–4.5
6–7	stationary-phase intuition	§4.6 (and §2.5)
7–11	Additional Note 1: canonical transformations, Hamilton–Jacobi, quantum analogue	§3.3–3.4, §4.2
12–14	derivation from Schrödinger theory; phase-space path integral	§5.1–5.2
15–17	Case 1, Case 2, conclusion	§5.3–5.5
18–20	propagator, retarded propagator, Green's function	§6
20–22	insertions, cutting and sewing, time ordering	§7.1–7.3
22–26	operator EOM; the oscillator both ways	§7.4–7.5
27–29	Additional Note 2: distributions	§2.4
29–31	functional derivative	§2.2–2.3
31–33	recap; issues in computing path integrals	§6–7, §8.1
34–36	Euclidean path integral, $i\varepsilon$ prescription	§8
37	measure; partition function	§9.1, §8.11
38–42	oscillator path integral; exercise	§9, App. A

B.2 Corrections

p. 2 "Minimization of S" $\to$ *stationarity*. The classical path is a minimum only before the first conjugate point (§3).

pp. 2–3, 9–11 The action functional $S[x]$ and Hamilton's principal function S_{cl} are now distinguished (§3.2). Only S_{cl} generates canonical transformations. On p. 10, $S([x], x', t) \to S_{\mathrm{cl}}(x, t; x', t_i)$.

p. 3 The argument that the formula "only makes sense classically" is refined: with S_{cl}, what fails is the prefactor, multiple classical paths, and finite

times (§4.3).

p. 4 Added: Way 1 is dimensionally inconsistent, so integrating over intermediate points is forced.

p. 6 "θ changes wildly between -1 and $+1$" $\to$ the real and imaginary parts of $e^{i\theta}$ do. Added: tube width $\sqrt{\hbar T/m}$; typical paths are not differentiable.

p. 8 Condition (3.7) is sufficient, not necessary. Only $\delta x = 0$ at the endpoints is needed.

p. 9 H' need only be a function of t, which can be absorbed into S.

p. 11 The operator form of Dirac's relations requires well-ordering, and $G = S_{\mathrm{cl}} + O(\hbar)$ is complex in general (Example 4.1).

p. 13 $H(p,x) \equiv \langle p|H|x\rangle / \langle p|x\rangle$ is an ordering prescription ($\hat{p}$ left of $\hat{x}$). Example: $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) \to px + i\hbar/2$. The midpoint rule corresponds to Weyl ordering.

p. 15 The constant "absorbed into the measure" is exactly A^N, which also fixes A.

p. 16 Missing factors of i and a sign in the expansion of $\mathcal{M}$ restored (5.12). The Jacobian $\hbar^{1/2}$ arises once per time slice. M_0 is a local, path-dependent measure factor (5.15), not a constant. The expansion carries $\delta(0)$ terms. The parity argument is now a proof.

p. 17 Added: the Legendre transform must exist, and it fails for constrained systems.

p. 19 Test functions are compactly supported, so no $f(\infty)$ terms arise.

p. 20 The margin question is answered (§6, Remark). Added: advanced propagator and energy representation.

p. 22 "Expectation value" $\to$ "matrix element".

p. 25 The missing $-e^{iS/\hbar}\delta(t-t')$ term is restored (7.12). The boundary question is answered in §8.10.

p. 27 "Given (A) we can find (B) but the reverse is not true" is corrected: for ordinary functions (B) determines (A); distributions are objects for which only (B) exists.

p. 28–29 The verification of the distributional identity used a test function constant in the integration variable, which proves nothing. Replaced by a proof with an arbitrary $\varphi(t)$ (Proposition 2.7).

p. 30 The functional derivative carries a factor $1/\varepsilon$; without it $\delta_{mm'} \not\to \delta(t-t')$.

p. 31 $\dot{x}(t) \to \dot{x}(t')$ inside $\int \mathrm{d}t'$.

p. 32 "All other matrix elements by time derivatives and coincident limits" is qualified by contact terms and discretization dependence (§7.7).

p. 33 $(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega) \to (\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega^2)$.

p. 34 "$E_n > 0$, bounded system" $\to \hat{H}$ bounded below. Added: why time ordering is needed.

p. 35 Signature noted: hw. uses $(-,+)$, Peskin uses $(+,-)$.

p. 36 $t = iu \to t = -iu$. "Damped away from its extremum" $\to$ dominated by minima of S_E. The $i\varepsilon$ term is shown to be equivalent to tilting time.

p. 37 $t = i\hbar\beta \to t = -i\hbar\beta$. The measure is fixed by the kinetic term alone. The real-time trace is not yet a partition function.

p. 39 Added: the Jacobian C is ω-independent, which is why the free particle fixes it. The real-time classical phase combined with a Euclidean fluctuation factor is justified.

p. 40 First line: $(1 - \omega^2/\lambda_n) \to (1 + \omega^2/\lambda_n)$ in Euclidean signature. $\frac{\sin x}{n} \to \frac{\sin x}{x}$. In $F(0)$, $(t_f - t_i) \to (t_f - t_i)^2$.

pp. 40–41 Added: the Maslov phase for $\omega T > \pi$, behaviour at the focal points, and the thermal-trace check of the spectrum.